Selected Exercises

Stochastic Methods

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Independently written solutions to selected exercises from
Introduction to Probability Models
by Sheldon M. Ross

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About these notes

These notes collect selected exercises and independently written solutions by Pietro Forni and Riccardo Visconti, prepared while studying stochastic methods. They are based on *Introduction to Probability Models* by Sheldon M. Ross and are unofficial study notes. No university or publisher endorsement is claimed.

Exercise numbers are retained from the original manuscript and restart within each topic. Cite a topic title and exercise number together. Some statements have been adapted or expanded with explicit assumptions so that the exercises can be read without the textbook. In particular, the two-ball urn and insurance premium models in Markov Chains Exercises 9 and 53 are fully specified versions chosen for these notes; they are not asserted to reproduce a particular textbook edition.

The collection has received a first mathematical review. Corrections and further improvements are welcome.

Conventions

Exponential and gamma distributions use rate parameters. Geometric random variables count trials through the first success and take values $1, 2, \ldots$, unless explicitly defined otherwise. Rates are positive. Conditioning formulas apply where the conditioning event has positive probability or the marginal density is positive; conditional expectations for continuous variables are defined almost everywhere. Variance identities assume finite second moments. Queueing models use independent service times and arrival processes unless stated otherwise.

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1 Probability

Exercise 1. A box contains three marbles: one red, one green, and one blue. Consider an experiment that consists of taking one marble from the box then replacing it in the box and drawing a second marble from the box. What is the sample space? If, at all times, each marble in the box is equally likely to be selected, what is the probability of each point in the sample space?

Exercise 2. Repeat Exercise 1 when the second marble is drawn without replacing the first marble.

Exercise 3. A coin is to be tossed until a head appears twice in a row. What is the sample space for this experiment? If the coin is fair, what is the probability that it will be tossed exactly four times?

Exercise 5. An individual uses the following gambling system at Las Vegas. He bets \$1 that the roulette wheel will come up red. If he wins, he quits. If he loses then he makes the same bet a second time only this time he bets \$2; and then regardless of the outcome, quits. Assuming that he has a probability of $\frac{1}{2}$ of winning each bet, what is the probability that he goes home a winner? Why is this system not used by everyone?

Exercise 11. If two fair dice are tossed, what is the probability that the sum is i, $i = 2, 3, \ldots, 12$?

Exercise 13. The dice game craps is played as follows. The player throws two dice, and if the sum is seven or eleven, then she wins. If the sum is two, three, or twelve, then she loses. If the sum is anything else, then she continues throwing until she either throws that number again (in which case she wins) or she throws a seven (in which case she loses). Calculate the probability that the player wins.

Exercise 14. The probability of winning on a single toss of the dice is p. A starts, and if he fails, he passes the dice to B, who then attempts to win on her toss. They continue tossing the dice back and forth until one of them wins. What are their respective probabilities of winning?

Exercise 17. Suppose each of three persons tosses a coin. If the outcome of one of the tosses differs from the other outcomes, then the game ends. If not, then the persons start over and retoss their coins. Assuming fair coins, what is the probability that the game will end with the first round of tosses? If all three coins are biased and have probability $\frac{1}{4}$ of landing heads, what is the probability that the game will end at the first round?

Exercise 18. Assume that each child who is born is equally likely to be a boy or a girl. If a family has two children, what is the probability that both are girls given that (a) the eldest is a girl, (b) at least one is a girl?

Exercise 19. Two dice are rolled. What is the probability that at least one is a six? If the two faces are different, what is the probability that at least one is a six?

Exercise 20. Three dice are thrown. What is the probability the same number appears on exactly two of the three dice?

Exercise 21. Suppose that 5 percent of men and 0.25 percent of women are color-blind. A color-blind person is chosen at random. What is the probability of this person being male? Assume that there are an equal number of males and females.

Exercise 22. A and B play until one has 2 more points than the other. Assuming that each point is independently won by A with probability p, what is the probability they will play a total of $2n$ points? What is the probability that A will win?

Exercise 25. Two cards are randomly selected from a deck of 52 playing cards.

(a) What is the probability they constitute a pair (that is, that they are of the same denomination)?

(b) What is the conditional probability they constitute a pair given that they are of different suits?

Exercise 26. A deck of 52 playing cards, containing all 4 aces, is randomly divided into 4 piles of 13 cards each. Define events E_1, E_2, E_3, and E_4 as follows:

$$\begin{aligned} E_1 &= \{\text{the first pile has exactly 1 ace}\},\\ E_2 &= \{\text{the second pile has exactly 1 ace}\},\\ E_3 &= \{\text{the third pile has exactly 1 ace}\},\\ E_4 &= \{\text{the fourth pile has exactly 1 ace}\} \end{aligned}$$

Find $P(E_1E_2E_3E_4)$, the probability that each pile has an ace.

Exercise 27. Suppose in Exercise 26 we had defined the events $E_i, i = 1, 2, 3, 4$, by

$E_1 = \{\text{one of the piles contains the ace of spades}\}$,
$E_2 = \{\text{the ace of spades and the ace of hearts are in different piles}\}$,
$E_3 = \{\text{the ace of spades, the ace of hearts, and the ace of diamonds are in different piles}\}$,
$E_4 = \{\text{all 4 aces are in different piles}\}$

Use the general multiplication rule for probabilities to find $P(E_1E_2E_3E_4)$, the probability that each pile has an ace. Compare your answer with the one you obtained in Exercise 26.

Exercise 30. Bill and George go target shooting together. Both shoot at a target at the same time. Suppose Bill hits the target with probability 0.7, whereas George, independently, hits the target with probability 0.4.

(a) Given that exactly one shot hit the target, what is the probability that it was George's shot?

(b) Given that the target is hit, what is the probability that George hit it?

Exercise 31. What is the conditional probability that the first die is six given that the sum of the dice is seven?

Exercise 32. Suppose all n men at a party throw their hats in the center of the room. Each man then randomly selects a hat. Show that the probability that none of the n men selects his own hat is

$$\frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \cdots + \frac{(-1)^n}{n!}$$

Note that as $n \to \infty$ this converges to e^{-1}. Is this surprising?

Exercise 33. In a class there are four freshman boys, six freshman girls, and six sophomore boys. How many sophomore girls must be present if sex and class are to be independent when a student is selected at random?

Exercise 34. Mr. Jones has devised a gambling system for winning at roulette. When he bets, he bets on red, and places a bet only when the ten previous spins of the roulette have landed on a black number. He reasons that his chance of winning is quite large since the probability of eleven consecutive spins resulting in black is quite small. What do you think of this system?

Exercise 35. A fair coin is continually flipped. What is the probability that the first four flips are

(a) H, H, H, H?

(b) T, H, H, H?

(c) What is the probability that the pattern T, H, H, H occurs before the pattern H, H, H, H?

Exercise 36. Consider two boxes, one containing one black and one white marble, the other, two black and one white marble. A box is selected at random and a marble is drawn at random from the selected box. What is the probability that the marble is black?

Exercise 37. In Exercise 36, what is the probability that the first box was the one selected given that the marble is white?

Exercise 38. Urn 1 contains two white balls and one black ball, while urn 2 contains one white ball and five black balls. One ball is drawn at random from urn 1 and placed in urn 2. A ball is then drawn from urn 2. It happens to be white. What is the probability that the transferred ball was white?

Exercise 39. Stores A, B, and C have 50, 75, and 100 employees, and, respectively, 50, 60, and 70 percent of these are women. Resignations are equally likely among all employees, regardless of sex. One employee resigns and this is a woman. What is the probability that she works in store C?

Exercise 40. (a) A gambler has in his pocket a fair coin and a two-headed coin. He selects one of the coins at random, and when he flips it, it shows heads. What is the probability that it is the fair coin?

(b) Suppose that he flips the same coin a second time and again it shows heads. Now what is the probability that it is the fair coin?

(c) Suppose that he flips the same coin a third time and it shows tails. Now what is the probability that it is the fair coin?

Exercise 41. In a certain species of rats, black dominates over brown. Suppose that a black rat with two black parents has a brown sibling.

(a) What is the probability that this rat is a pure black rat (as opposed to being a hybrid with one black and one brown gene)?

(b) Suppose that when the black rat is mated with a brown rat, all five of their offspring are black. Now, what is the probability that the rat is a pure black rat?

Exercise 42. There are three coins in a box. One is a two-headed coin, another is a fair coin, and the third is a biased coin that comes up heads 75 percent of the time. When one of the three coins is selected at random and flipped, it shows heads. What is the probability that it was the two-headed coin?

Exercise 43. Suppose we have ten coins which are such that if the ith one is flipped then heads will appear with probability $i/10, i = 1, 2, \ldots, 10$. When one of the coins is randomly selected

and flipped, it shows heads. What is the conditional probability that it was the fifth coin?

Exercise 44. Urn 1 has five white and seven black balls. Urn 2 has three white and twelve black balls. We flip a fair coin. If the outcome is heads, then a ball from urn 1 is selected, while if the outcome is tails, then a ball from urn 2 is selected. Suppose that a white ball is selected. What is the probability that the coin landed tails?

Exercise 45. An urn contains b black balls and r red balls. One of the balls is drawn at random, but when it is put back in the urn c additional balls of the same color are put in with it. Now suppose that we draw another ball. Show that the probability that the first ball drawn was black given that the second ball drawn was red is $b/(b+r+c)$.

Exercise 46. Three prisoners are informed by their jailer that one of them has been chosen at random to be executed, and the other two are to be freed. Prisoner A asks the jailer to tell him privately which of his fellow prisoners will be set free, claiming that there would be no harm in divulging this information, since he already knows that at least one will go free. The jailer refuses to answer this question, pointing out that if A knew which of his fellows were to be set free, then his own probability of being executed would rise from $\frac{1}{3}$ to $\frac{1}{2}$, since he would then be one of two prisoners. What do you think of the jailer's reasoning?

Exercise 48. Sixty percent of the families in a certain community own their own car, thirty percent own their own home, and twenty percent own both their own car and their own home. If a family is randomly chosen, what is the probability that this family owns a car or a house but not both?

2 Random Variables

Exercise 1. An urn contains five red, three orange, and two blue balls. Two balls are randomly selected. What is the sample space of this experiment? Let X represent the number of orange balls selected. What are the possible values of X? Calculate $P\{X = 0\}$.

Exercise 2. Let X represent the difference between the number of heads and the number of tails obtained when a coin is tossed n times. What are the possible values of X?

Exercise 3. In Exercise 2, if the coin is assumed fair, then, for $n = 2$, what are the probabilities associated with the values that X can take on?

Exercise 4. Suppose a die is rolled twice. What are the possible values that the following random variables can take on?

(a) The maximum value to appear in the two rolls.

(b) The minimum value to appear in the two rolls.

(c) The sum of the two rolls.

(d) The value of the first roll minus the value of the second roll.

Exercise 5. If the die in Exercise 4 is assumed fair, calculate the probabilities associated with the random variables in (i)–(iv).

Exercise 6. Suppose five fair coins are tossed. Let E be the event that all coins land heads. Define the random variable I_E by

$$I_E = \begin{cases} 1, & \text{if } E \text{ occurs,} \\ 0, & \text{if } E^c \text{ occurs.} \end{cases}$$

For what outcomes in the original sample space does I_E equal 1? What is $P\{I_E = 1\}$?

Exercise 7. Suppose a coin having probability 0.7 of coming up heads is tossed three times. Let X denote the number of heads that appear in the three tosses. Determine the probability mass function of X.

Exercise 8. Suppose the distribution function of X is given by

$$F(b) = \begin{cases} 0, & b < 0, \\ \frac{1}{2}, & 0 \le b < 1, \\ 1, & b \ge 1. \end{cases}$$

What is the probability mass function of X?

Exercise 9. If the distribution function of X is given by

$$F(b) = \begin{cases} 0, & b < 0, \\ \frac{1}{2}, & 0 \le b < 1, \\ \frac{3}{5}, & 1 \le b < 2, \\ \frac{4}{5}, & 2 \le b < 3, \\ \frac{9}{10}, & 3 \le b < 3.5, \\ 1, & b \ge 3.5, \end{cases}$$

calculate the probability mass function of X.

Exercise 10. Suppose three fair dice are rolled. What is the probability that at most one six appears?

Exercise 11. A ball is drawn from an urn containing three white and three black balls. After the ball is drawn, it is replaced and another ball is drawn. This goes on indefinitely. What is the probability that of the first four balls drawn, exactly two are white?

Exercise 12. On a multiple-choice exam with three possible answers for each of the five questions, what is the probability that a student would get four or more correct answers just by guessing?

Exercise 13. An individual claims to have extrasensory perception (ESP). As a test, a fair coin is flipped ten times, and he is asked to predict in advance the outcome. Our individual gets seven out of ten correct. What is the probability he would have done at least this well if he had no ESP?

Exercise 14. Suppose X has a binomial distribution with parameters 6 and $\frac{1}{2}$. Show that $X = 3$ is the most likely outcome.

Exercise 15. Let X be binomially distributed with parameters n and p. Show that as k goes from 0 to n, $P(X = k)$ increases monotonically, then decreases monotonically, reaching its largest value

(a) when $(n+1)p$ is an integer, at $k = (n+1)p - 1$ and $k = (n+1)p$,

(b) when $(n+1)p$ is not an integer, at the unique integer k satisfying

$$(n+1)p - 1 < k < (n+1)p.$$

Exercise 16. An airline knows that 5 percent of the people making reservations on a certain flight will not show up. Consequently, their policy is to sell 52 tickets for a flight that can hold only 50 passengers. What is the probability that there will be a seat available for every passenger who shows up?

Exercise 17. Suppose that an experiment can result in one of r possible outcomes, the ith outcome having probability p_i, $i = 1, \ldots, r$, with

$$\sum_{i=1}^{r} p_i = 1.$$

If n of these experiments are performed independently, show that the probability that the first outcome appears x_1 times, the second x_2 times, and the rth x_r times is

$$\frac{n!}{x_1! x_2! \cdots x_r!} p_1^{x_1} p_2^{x_2} \cdots p_r^{x_r},$$

where

$$x_1 + x_2 + \cdots + x_r = n.$$

Exercise 18. Show that when $r = 2$ the multinomial reduces to the binomial.

Exercise 19. In Exercise 17, let X_i denote the number of times the ith outcome appears, $i = 1, \ldots, r$. What is the probability mass function of

$$X_1 + X_2 + \cdots + X_k?$$

Exercise 20. A television store owner figures that 50 percent of the customers entering his store will purchase an ordinary television set, 20 percent will purchase a color television set, and 30 percent will just be browsing. If five customers enter his store on a certain day, what is the probability that two customers purchase color sets, one customer purchases an ordinary set, and two customers purchase nothing?

Exercise 21. In Exercise 20, what is the probability that our store owner sells three or more televisions on that day?

Exercise 22. If a fair coin is successively flipped, find the probability that a head first appears on the fifth trial.

Exercise 23. A coin having probability p of coming up heads is successively flipped until the rth head appears. Argue that X, the number of flips required, will be n, $n \geq r$, with probability

$$P(X = n) = \binom{n-1}{r-1} p^r (1-p)^{n-r}.$$

Exercise 24. The probability mass function of X is given by

$$p(k) = \binom{r+k-1}{r-1} p^r (1-p)^k, \quad k = 0, 1, \ldots$$

Give a possible interpretation of the random variable X.

In Exercises 25 and 26, suppose that two teams are playing a series of games, each of which is independently won by team A with probability p and by team B with probability $1 - p$. The winner of the series is the first team to win i games.

Exercise 25. If $i = 4$, find the probability that a total of 7 games are played. Also show that this probability is maximized when $p = \frac{1}{2}$.

Exercise 26. Find the expected number of games that are played when

(a) $i = 2$;

(b) $i = 3$.

In both cases, show that this number is maximized when $p = \frac{1}{2}$.

Exercise 27. A fair coin is independently flipped n times, k times by A and $n - k$ times by B. Show that the probability that A and B flip the same number of heads is equal to the probability that there are a total of k heads.

Exercise 28. Suppose that we want to generate a random variable X equally likely to be either 0 or 1, and all we have at our disposal is a biased coin with (unknown) probability p of landing heads. Consider the following procedure:

(a) Flip the coin, and let O_1 be the result.

(b) Flip the coin again, and let O_2 be the result.

(c) If O_1 and O_2 are the same, return to step 1.

(d) If O_2 is heads, set $X = 0$; otherwise set $X = 1$.

(a) Show that X is equally likely to be either 0 or 1.

(b) Could we use a simpler procedure that continues to flip the coin until the last two flips are different, and then sets $X = 0$ if the final flip is a head, and $X = 1$ if it is a tail?

Exercise 29. Consider n independent flips of a coin having probability p of landing heads. Say a *changeover* occurs whenever an outcome differs from the one preceding it. If $p = \frac{1}{2}$, what is the probability there are k changeovers?

Exercise 30. Let X be a Poisson random variable with parameter λ. Show that $P\{X = i\}$ increases monotonically and then decreases monotonically as i increases, reaching its maximum when i is the largest integer not exceeding λ.

Exercise 31. Compare the Poisson approximation with the correct binomial probability for the following cases:

(a) $P\{X = 2\}$ when $n = 8$, $p = 0.1$.

(b) $P\{X = 9\}$ when $n = 10$, $p = 0.95$.

(c) $P\{X = 0\}$ when $n = 10$, $p = 0.1$.

(d) $P\{X = 4\}$ when $n = 9$, $p = 0.2$.

Exercise 32. If you buy a lottery ticket in 50 lotteries, in each of which your chance of winning a prize is $\frac{1}{100}$, what is the (approximate) probability that you will win a prize (a) at least once, (b) exactly once, (c) at least twice?

Exercise 33. Let X be a random variable with probability density

$$f(x) = \begin{cases} c(1 - x^2), & -1 < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

(a) What is the value of c?

(b) What is the cumulative distribution function of X?

Exercise 34. Let the probability density of X be given by

$$f(x) = \begin{cases} c(4x - 2x^2), & 0 < x < 2, \\ 0, & \text{otherwise.} \end{cases}$$

(a) What is the value of c?

(b) $P\left\{\frac{1}{2} < X < \frac{3}{2}\right\} = ?$

Exercise 35. The density of X is given by

$$f(x) = \begin{cases} 10/x^2, & x > 10, \\ 0, & x \leq 10. \end{cases}$$

What is the distribution of X? Find $P\{X > 20\}$.

Exercise 36. A point is uniformly distributed within the disk of radius 1. That is, its density is

$$f(x, y) = C, \quad 0 \leq x^2 + y^2 \leq 1.$$

Find the probability that its distance from the origin is less than x, $0 \leq x \leq 1$.

Exercise 37. Let $X_1, X_2, \ldots, X_n$ be independent random variables, each having a uniform distribution over $(0, 1)$. Let $M = \max(X_1, \ldots, X_n)$. Show that the distribution function of M is given by $F_M(x) = x^n$ for $0 \leq x \leq 1$. What is the probability density function of M?

Exercise 38. If the density function of X equals

$$f(x) = \begin{cases} ce^{-2x}, & 0 < x < \infty, \\ 0, & x < 0, \end{cases}$$

find c. What is $P\{X > 2\}$?

Exercise 39. The random variable X has the following probability mass function:

$$p(1) = \tfrac{1}{2}, \quad p(2) = \tfrac{1}{3}, \quad p(24) = \tfrac{1}{6}.$$

Calculate $E[X]$.

Exercise 40. Suppose that two teams are playing a series of games, each of which is independently won by team A with probability p and by team B with probability $1 - p$. The winner of the series is the first team to win four games. Find the expected number of games that are played, and evaluate this quantity when $p = \frac{1}{2}$.

Exercise 41. Consider the case of arbitrary p in Exercise 29. Compute the expected number of changeovers.

Exercise 42. Suppose that each coupon obtained is, independent of what has been previously obtained, equally likely to be any of m different types. Find the expected number of coupons one needs to obtain in order to have at least one of each type.

Exercise 43. An urn contains $n + m$ balls, of which n are red and m are black. They are withdrawn one at a time without replacement. Let X be the number of red balls removed before the first black ball is chosen.

(a) Express X in terms of indicator random variables X_i, $i = 1, \ldots, n$, where $X_i = 1$ if red ball i is taken before any black ball.

(b) Find $E[X]$.

Exercise 44. In Exercise 43, let Y denote the number of red balls chosen after the first but before the second black ball has been chosen.

(a) Express Y as the sum of n indicator random variables.

(b) Find $E[Y]$.

(c) Compare $E[Y]$ to $E[X]$ obtained in Exercise 43.

(d) Can you explain the result obtained in (c)?

Exercise 45. A total of r keys are to be put, one at a time, in k boxes, with each key independently being put in box i with probability p_i, $\sum_{i=1}^{k} p_i = 1$. Each time a key is put in a nonempty box, a collision occurs. Find the expected number of collisions.

Exercise 46. If X is a nonnegative integer-valued random variable, show that

$$E[X] = \sum_{n=1}^{\infty} P\{X \geq n\} = \sum_{n=0}^{\infty} P\{X > n\}.$$

Also, if X and Y are both nonnegative integer-valued, show that

$$E[XY] = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} P(X \geq n, Y \geq m).$$

Exercise 47. Consider three trials, each of which is either a success or not. Let X denote the number of successes. Suppose that $E[X] = 1.8$.

(a) What is the largest possible value of $P\{X = 3\}$?

(b) What is the smallest possible value of $P\{X = 3\}$?

In both cases, construct a probability scenario that results in $P\{X = 3\}$ having the desired value.

Exercise 48. If X is a nonnegative random variable and g is a differentiable function with $g(0) = 0$, then

$$E[g(X)] = \int_0^{\infty} P(X > t)\, g'(t)\, dt.$$

Prove the above when X is a continuous random variable.

Exercise 49. Prove that $E[X^2] \geq (E[X])^2$. When do we have equality?

Exercise 50. Let c be a constant. Show that

(a) $\mathrm{Var}(cX) = c^2 \mathrm{Var}(X)$;

(b) $\mathrm{Var}(c + X) = \mathrm{Var}(X)$.

Exercise 51. A coin having probability p of landing heads is flipped until a head appears for the r-th time. Let N denote the number of flips required. Calculate $E[N]$.

Exercise 52. (a) Calculate $E[X]$ for the maximum random variable M of Exercise 37.

(b) Calculate $E[X]$ for X as in Exercise 33.

(c) Calculate $E[X]$ for X as in Exercise 34.

Exercise 53. If X is uniform over $(0, 1)$, calculate $E[X^n]$ and $\mathrm{Var}(X^n)$.

Exercise 54. Let X and Y each take on either the value 1 or -1. Let $p(1, 1)$, $p(1, -1)$, $p(-1, 1)$, $p(-1, -1)$ be the joint probabilities. Suppose $E[X] = E[Y] = 0$. Show that:

(a) $p(1, 1) = p(-1, -1)$;

(b) $p(1, -1) = p(-1, 1)$.

Let $p = 2p(1, 1)$. Find (c) $\mathrm{Var}(X)$; (d) $\mathrm{Var}(Y)$; (e) $\mathrm{Cov}(X, Y)$.

Exercise 55. Suppose that the joint probability mass function of X and Y is

$$P(X = i, Y = j) = \binom{j}{i} e^{-2\lambda} \frac{\lambda^j}{j!}, \quad 0 \leq i \leq j, \quad j = 0, 1, 2, \ldots$$

(a) Find the probability mass function of Y.

(b) Find the probability mass function of X.

(c) Find the probability mass function of $Y - X$.

Exercise 56. There are n types of coupons. Each newly obtained coupon is, independently, type i with probability p_i, $i = 1, \ldots, n$. Find the expected number and the variance of the number of distinct types obtained in a collection of k coupons.

Exercise 57. Suppose that X and Y are independent binomial random variables with parameters (n, p) and (m, p), respectively. Argue probabilistically (no computations necessary) that $X + Y$ is binomial with parameters $(n + m, p)$.

Exercise 58. An urn contains $2n$ balls, of which r are red. The balls are randomly removed in n successive pairs. Let X denote the number of pairs in which both balls are red.

(a) Find $E[X]$.

(b) Find $\mathrm{Var}(X)$.

Exercise 59. Let X_1, X_2, X_3, X_4 be independent continuous random variables with a common distribution function F and let

$$p = P\{X_1 < X_2 > X_3 < X_4\}.$$

(a) Argue that the value of p is the same for all continuous distribution functions F.

(b) Find p by integrating the joint density function over the appropriate region.

(c) Find p by using the fact that all 4! possible orderings of $X_1, \ldots, X_4$ are equally likely.

Exercise 60. Calculate the moment generating function of the uniform distribution on $(0, 1)$. Obtain $E[X]$ and $\mathrm{Var}[X]$ by differentiating.

Exercise 61. Let X and W be the working and subsequent repair times of a certain machine. Let $Y = X + W$ and suppose the joint density of X and Y is

$$f_{X,Y}(x, y) = \lambda^2 e^{-\lambda y}, \quad 0 < x < y < \infty.$$

(a) Find the density of X.

(b) Find the density of Y.

(c) Find the joint density of X and W.

(d) Find the density of W.

Exercise 62. Insurance companies sometimes use the exponential principle: with X the random amount to be paid in claims, the premium is $P = \frac{1}{a} \ln E[e^{aX}]$. Find P when X is exponential with parameter λ, and $a = \alpha\lambda$ where $0 < \alpha < 1$.

Exercise 63. Calculate the moment generating function of a geometric random variable.

Exercise 64. Show that the sum of independent identically distributed exponential random variables has a gamma distribution.

Exercise 65. Let $Z_1, \ldots, Z_n$ be independent standard normal random variables. For fixed real coefficients a_{rs} and constants μ_r, define $X_r = \mu_r + \sum_{s=1}^{n} a_{rs} Z_s$. Find $\mathrm{Cov}(X_i, X_j)$ in terms of the coefficients.

Exercise 66. Use Chebyshev's inequality to prove the weak law of large numbers: if $X_1, X_2, \ldots$ are i.i.d. with mean μ and variance σ^2, then for any $\varepsilon > 0$,

$$P\left\{\left|\frac{X_1 + \cdots + X_n}{n} - \mu\right| > \varepsilon\right\} \to 0 \quad \text{as } n \to \infty.$$

Exercise 67. Suppose that X is a random variable with mean 10 and variance 15. What can we say about $P\{5 < X < 15\}$?

Exercise 68. Let $X_1, X_2, \ldots, X_{10}$ be independent Poisson random variables with mean 1.

(a) Use the Markov inequality to obtain a bound on $P\{X_1 + \cdots + X_{10} \geq 15\}$.

(b) Use the central limit theorem to approximate $P\{X_1 + \cdots + X_{10} \geq 15\}$.

Exercise 69. If X is normally distributed with mean 1 and variance 4, use the tables to find $P\{2 < X < 3\}$.

Exercise 70. Show that

$$\lim_{n\to\infty} e^{-n} \sum_{k=0}^{n} \frac{n^k}{k!} = \frac{1}{2}.$$

Exercise 71. Let X denote the number of white balls selected when k balls are chosen at random from an urn containing n white and m black balls.

(a) Compute $P\{X = i\}$.

(b) Let $X_i = \mathbf{1}[i\text{-th ball selected is white}]$ and $Y_j = \mathbf{1}[\text{white ball } j \text{ is selected}]$. Compute $E[X]$ using first X_i's then Y_j's.

Exercise 72. Show that $\mathrm{Var}(X) = 1$ when X is the number of men who select their own hats (as in the hat-matching problem with n men).

Exercise 73. For the multinomial distribution (Exercise 17), let N_i denote the number of times outcome i occurs. Find:

(a) $E[N_i]$;

(b) $\mathrm{Var}(N_i)$;

(c) $\mathrm{Cov}(N_i, N_j)$ for $i \neq j$;

(d) the expected number of outcomes that do not occur.

3 Conditioning and Expectation

Exercise 1. If X and Y are both discrete, show that $\sum_x p_{X|Y}(x|y) = 1$ for all y such that $p_Y(y) > 0$.

Exercise 2. Let X_1 and X_2 be independent geometric random variables having the same parameter p. Guess the value of $P\{X_1 = i \mid X_1 + X_2 = n\}$.

Hint: Suppose a coin having probability p of coming up heads is continually flipped. If the second head occurs on flip number n, what is the conditional probability that the first head was on flip number i, $i = 1, \ldots, n-1$? Verify your guess analytically.

Exercise 3. The joint probability mass function of X and Y, $p(x, y)$, is given by

$$\begin{aligned} p(1,1) &= \frac{1}{9}, & p(2,1) &= \frac{1}{3}, & p(3,1) &= \frac{1}{9}, \\ p(1,2) &= \frac{1}{9}, & p(2,2) &= 0, & p(3,2) &= \frac{1}{18}, \\ p(1,3) &= 0, & p(2,3) &= \frac{1}{6}, & p(3,3) &= \frac{1}{9} \end{aligned}$$

Compute $E[X \mid Y = i]$ for $i = 1, 2, 3$.

Exercise 4. In Exercise 3, are the random variables X and Y independent?

Exercise 5. An urn contains three white, six red, and five black balls. Six of these balls are randomly selected from the urn. Let X and Y denote respectively the number of white and black balls selected. Compute the conditional probability mass function of X given that $Y = 3$. Also compute $E[X \mid Y = 1]$.

Exercise 6. Repeat Exercise 5 but under the assumption that when a ball is selected its color is noted, and it is then replaced in the urn before the next selection is made.

Exercise 7. Suppose $p(x, y, z)$, the joint probability mass function of the random variables X, Y, and Z, is given by

$$\begin{aligned} p(1,1,1) &= \frac{1}{8}, & p(2,1,1) &= \frac{1}{4}, \\ p(1,1,2) &= \frac{1}{8}, & p(2,1,2) &= \frac{3}{16}, \\ p(1,2,1) &= \frac{1}{16}, & p(2,2,1) &= 0, \\ p(1,2,2) &= 0, & p(2,2,2) &= \frac{1}{4} \end{aligned}$$

What is $E[X \mid Y = 2]$? What is $E[X \mid Y = 2, Z = 1]$?

Exercise 8. An unbiased die is successively rolled. Let X and Y denote, respectively, the number of rolls necessary to obtain a six and a five. Find (a) $E[X]$, (b) $E[X \mid Y = 1]$, (c) $E[X \mid Y = 5]$.

Exercise 9. Show in the discrete case that if X and Y are independent, then $E[X \mid Y = y] = E[X]$ for all y.

Exercise 10. Suppose X and Y are independent continuous random variables. Show that $E[X \mid Y = y] = E[X]$ for all y.

Exercise 11. The joint density of X and Y is

$$f(x,y) = \frac{y^2 - x^2}{8} e^{-y}, \quad 0 < y < \infty, \quad -y \le x \le y$$

Show that $E[X \mid Y = y] = 0$.

Exercise 12. The joint density of X and Y is given by

$$f(x,y) = \frac{e^{-x/y} e^{-y}}{y}, \quad 0 < x < \infty, \quad 0 < y < \infty$$

Show $E[X \mid Y = y] = y$.

Exercise 13. Let X be exponential with mean $1/\lambda$; that is,

$$f_X(x) = \lambda e^{-\lambda x}, \quad 0 < x < \infty$$

Find $E[X \mid X > 1]$.

Exercise 14. Let X be uniform over $(0,1)$. Find $E\left[X \mid X < \frac{1}{2}\right]$.

Exercise 15. The joint density of X and Y is given by

$$f(x,y) = \frac{e^{-y}}{y}, \quad 0 < x < y, \quad 0 < y < \infty$$

Compute $E[X^2 \mid Y = y]$.

Exercise 19. Prove that if X and Y are jointly continuous, then

$$E[X] = \int_{-\infty}^{\infty} E[X \mid Y = y] f_Y(y)\, dy$$

Exercise 20. An individual whose level of exposure to a certain pathogen is x will contract the disease caused by this pathogen with probability $P(x)$. If the exposure level of a randomly chosen member of the population has probability density function f, determine the conditional probability density of the exposure level of that member given that he or she

(a) has the disease.

(b) does not have the disease.

(c) Show that when $P(x)$ increases in x, then the ratio of the density of part (a) to that of part (b) also increases in x.

Exercise 21. A miner is trapped in a mine with three doors. Door 1 leads to safety in 2 hours; doors 2 and 3 return the miner to the starting point after 3 and 5 hours, respectively. At each return, the miner independently chooses each door with probability 1/3. Let N denote the total number of doors selected before the miner reaches safety. Also, let T_i denote the travel time corresponding to the ith choice, $i \ge 1$. Let X denote the time when the miner reaches safety.

(a) Give an identity that relates X to N and the T_i.

(b) What is $E[N]$?

(c) What is $E[T_N]$?

(d) What is $E[\sum_{i=1}^{N} T_i \mid N = n]$?

(e) Using the preceding, what is $E[X]$?

Exercise 23. A coin having probability p of coming up heads is successively flipped until two of the most recent three flips are heads. Let N denote the number of flips. (Note that if the first two flips are heads, then $N = 2$.) Find $E[N]$.

Exercise 24. A coin, having probability p of landing heads, is continually flipped until at least one head and one tail have been flipped.

(a) Find the expected number of flips needed.

(b) Find the expected number of flips that land on heads.

(c) Find the expected number of flips that land on tails.

(d) Repeat part (a) in the case where flipping is continued until a total of at least two heads and one tail have been flipped.

Exercise 25. Independent trials, resulting in one of the outcomes 1, 2, 3 with respective probabilities p_1, p_2, p_3, $\sum_{i=1}^{3} p_i = 1$, are performed.

(a) Let N denote the number of trials needed until the initial outcome has occurred exactly 3 times. Find $E[N]$.

(b) Find the expected number of trials needed until both outcome 1 and outcome 2 have occurred.

Exercise 26. You have two opponents with whom you alternate play. Whenever you play A, you win with probability p_A; whenever you play B, you win with probability p_B, where $p_B > p_A$. If your objective is to minimize the expected number of games you need to play to win two in a row, should you start with A or with B?

Exercise 27. A coin that comes up heads with probability p is continually flipped until the pattern T, T, H appears. Let X denote the number of flips made, and find $E[X]$.

Exercise 28. Polya's urn model supposes that an urn initially contains r red and b blue balls. At each stage a ball is randomly selected from the urn and is then returned along with m other balls of the same color. Let X_k be the number of red balls drawn in the first k selections. Find (a) $E[X_1]$, (b) $E[X_2]$, (c) $E[X_3]$. (d) Conjecture the value of $E[X_k]$ and verify it by conditioning. (e) Give an intuitive proof.

Exercise 29. Two players take turns shooting at a target, with each shot by player i hitting the target with probability p_i, $i = 1, 2$. Shooting ends when two consecutive shots hit the target. Let μ_i denote the mean number of shots taken when player i shoots first, $i = 1, 2$. Find (a) μ_1 and μ_2, (b) h_1 and h_2 (the mean number of hits).

Exercise 30. Let $X_i, i \geq 0$ be independent and identically distributed random variables with probability mass function $p(j) = P\{X_i = j\}, j = 1, \ldots, m$. Find $E[N]$, where $N = \min\{n > 0 : X_n = X_0\}$.

Exercise 31. Each element in a sequence of binary data is either 1 with probability p or 0 with probability $1 - p$. A maximal subsequence of consecutive values having identical outcomes is called a run. Find (a) the expected length of the first run, (b) the expected length of the second run.

Exercise 32. Independent trials, each resulting in success with probability p, are performed.

(a) Find the expected number of trials needed for there to have been both at least n successes and at least m failures.

(b) Find the expected number of trials needed for there to have been either at least n successes or at least m failures.

Exercise 33. If R_i denotes the random amount that is earned in period i, then $\sum_{i=1}^{\infty} \beta^{i-1} R_i$, where $0 < \beta < 1$ is a specified constant, is called the total discounted reward with discount factor β. Let T be a geometric random variable with parameter $1 - \beta$ that is independent of the R_i. Show that the expected total discounted reward is equal to the expected total (undiscounted) reward earned by time T. That is, show that

$$E\left[\sum_{i=1}^{\infty} \beta^{i-1} R_i\right] = E\left[\sum_{i=1}^{T} R_i\right]$$

Exercise 34. A set of n dice is thrown. All those that land on six are put aside, and the others are again thrown. This is repeated until all the dice have landed on six. Let N denote the number of throws needed. Let $m_n = E[N]$.

(a) Derive a recursive formula for m_n and use it to calculate $m_i, i = 2, 3, 4$ and to show that $m_5 \approx 13.024$.

(b) Let X_i denote the number of dice rolled on the ith throw. Find $E\left[\sum_{i=1}^{N} X_i\right]$.

Exercise 35. Consider n multinomial trials, where each trial independently results in outcome i with probability p_i, $\sum_{i=1}^{k} p_i = 1$. With X_i equal to the number of trials that result in outcome i, find $E[X_1 \mid X_2 > 0]$.

Exercise 36. Let $p_0 = P\{X = 0\}$ and suppose that $0 < p_0 < 1$. Let $\mu = E[X]$ and $\sigma^2 = \text{Var}(X)$.

(a) Find $E[X \mid X \neq 0]$.

(b) Find $\text{Var}(X \mid X \neq 0)$.

Exercise 37. A manuscript is sent to a typing firm consisting of typists A, B, and C. If it is typed by A, then the number of errors made is a Poisson random variable with mean 2.6; if typed by B, then the number of errors is a Poisson random variable with mean 3; and if typed by C, then it is a Poisson random variable with mean 3.4. Let X denote the number of errors in the typed manuscript. Assume that each typist is equally likely to do the work. Find (a) $E[X]$, (b) $\text{Var}(X)$.

Exercise 38. Let U be a uniform $(0, 1)$ random variable. Suppose that n trials are to be performed and that conditional on $U = u$ these trials will be independent with a common success probability u. Compute the mean and variance of the number of successes that occur in these trials.

Exercise 39. A deck of n cards, numbered 1 through n, is randomly shuffled so that all $n!$ possible permutations are equally likely. The cards are then turned over one at a time until card number 1 appears. These upturned cards constitute the first cycle. We now determine (by looking at the upturned cards) the lowest numbered card that has not yet appeared, and we continue to turn the cards face up until that card appears. This new set of cards represents the second cycle. We again determine the lowest numbered of the remaining cards and turn the cards until it appears, and so on until all cards have been turned over. Let m_n denote the mean number of cycles.

(a) Derive a recursive formula for m_n in terms of $m_k, k = 1, \ldots, n-1$.

(b) Starting with $m_0 = 0$, find m_1, m_2, m_3, and m_4.

(c) Conjecture a general formula for m_n.

(d) Prove your formula by induction on n.

(e) Let $X_i = 1$ if a cycle ends with card i, 0 otherwise. Express cycles using X_i.

(f) Use part (e) to determine m_n.

(g) Are the random variables $X_1, \ldots, X_n$ independent? Explain.

(h) Find the variance of the number of cycles.

Exercise 40. A prisoner is trapped in a cell containing three doors. The first door leads to a tunnel that returns him to his cell after two days of travel. The second leads to a tunnel that returns him to his cell after three days of travel. The third door leads immediately to freedom.

(a) Assuming that the prisoner will always select doors 1, 2, and 3 with prob- abilities 0.5, 0.3, 0.2, what is the expected number of days until he reaches freedom?

(b) Assuming that the prisoner is always equally likely to choose among those doors that he has not used, what is the expected number of days until he reaches freedom? (In this version, for instance, if the prisoner initially tries door 1, then when he returns to the cell, he will now select only from doors 2 and 3.)

(c) For parts (a) and (b) find the variance of the number of days until the prisoner reaches freedom.

Exercise 41. A rat is trapped in a maze. Initially it has to choose one of two directions. If it goes to the right, then it will wander around in the maze for three minutes and will then return to its initial position. If it goes to the left, then with probability $\frac{1}{3}$ it will depart the maze after two minutes of traveling, and with probability $\frac{2}{3}$ it will return to its initial position after five minutes of traveling. Assuming that the rat is at all times equally likely to go to the left or the right, what is the expected number of minutes that it will be trapped in the maze?

Exercise 42. If X_i, $i = 1, \ldots, n$, are independent normal random variables, with X_i having mean μ_i and variance 1, then the random variable

$$\sum_{i=1}^{n} X_i^2$$

is said to be a *noncentral chi-squared* random variable.

(a) If X is a normal random variable having mean μ and variance 1, show that, for $|t| < \frac{1}{2}$, the moment generating function of X^2 is

$$M_{X^2}(t) = (1-2t)^{-1/2} \exp\left(\frac{t\mu^2}{1-2t}\right).$$

(b) Derive the moment generating function of the noncentral chi-squared random variable

$$\sum_{i=1}^{n} X_i^2,$$

and show that its distribution depends on the sequence of means $\mu_1, \ldots, \mu_n$ only through the sum of their squares. As a result, show that

$$\sum_{i=1}^{n} X_i^2$$

is a noncentral chi-squared random variable with parameters n and $\theta = \sum_{i=1}^{n} \mu_i^2$.

(c) If all $\mu_i = 0$, then

$$\sum_{i=1}^{n} X_i^2$$

is called a chi-squared random variable with n degrees of freedom. Determine, by differentiating its moment generating function, its expected value and variance.

(d) Let K be a Poisson random variable with mean $\theta/2$, and suppose that, conditional on $K = k$, the random variable W has a chi-squared distribution with $n + 2k$ degrees of freedom. Show, by computing its moment generating function, that W is a noncentral chi-squared random variable with parameters n and θ.

(e) Find the expected value and variance of a noncentral chi-squared random variable with parameters n and θ.

Exercise 43. The density function of a chi-squared random variable having n degrees of freedom can be shown to be

$$f(x) = \frac{\frac{1}{2}e^{-x/2}(x/2)^{n/2-1}}{\Gamma(n/2)}, \qquad x > 0,$$

where the gamma function is defined by

$$\Gamma(t) = \int_0^\infty e^{-x} x^{t-1}\, dx, \qquad t > 0.$$

Integration by parts can be employed to show that

$$\Gamma(t) = (t-1)\Gamma(t-1), \qquad t > 1.$$

If Z and χ_n^2 are independent random variables with Z having a standard normal distribution and χ_n^2 having a chi-squared distribution with n degrees of freedom, then the random variable

$$T = \frac{Z}{\sqrt{\chi_n^2/n}}$$

is said to have a t-distribution with n degrees of freedom.

Compute the mean and variance of T when $n > 2$.

Exercise 46. (a) Show that

$$\mathrm{Cov}(X, Y) = \mathrm{Cov}(X, E[Y \mid X])$$

(b) Suppose that, for constants a and b,

$$E[Y \mid X] = a + bX$$

Show that

$$b = \mathrm{Cov}(X, Y)/\mathrm{Var}(X)$$

Exercise 47. If $E[Y \mid X] = 1$, show that

$$\text{Var}(XY) \geqslant \text{Var}(X)$$

Exercise 51. If X is geometric with parameter p, find the probability that X is even.

Exercise 52. Suppose that X and Y are independent random variables with probability density functions f_X and f_Y. Determine a one-dimensional integral expression for $P\{X + Y < x\}$.

Exercise 53. *Suppose X is a Poisson random variable with mean λ. The parameter λ is itself a random variable whose distribution is exponential with mean 1. Show that $P\{X = n\} = \left(\frac{1}{2}\right)^{n+1}$.

Exercise 54. A coin is randomly selected from a group of ten coins, the nth coin having a probability $n/10$ of coming up heads. The coin is then repeatedly flipped until a head appears. Let N denote the number of flips necessary. What is the probability distribution of N? Is N a geometric random variable? When would N be a geometric random variable; that is, what would have to be done differently?

Exercise 55. You are invited to a party. Suppose the times at which invitees arrive are independent uniform $(0, 1)$ random variables. Suppose that, aside from yourself, the number of other people who are invited is a Poisson random variable with mean 10. (a) Find the expected number of people who arrive before you. (b) Find the probability that you are the nth person to arrive.

Exercise 56. Data indicate that the number of traffic accidents in Berkeley on a rainy day is a Poisson random variable with mean 9, whereas on a dry day it is a Poisson random variable with mean 3. Let X denote the number of traffic accidents tomorrow. If it will rain tomorrow with probability 0.6, find (a) $E[X]$; (b) $P\{X = 0\}$; (c) $\text{Var}(X)$.

Exercise 57. The number of storms in the upcoming rainy season is Poisson distributed but with a parameter value that is uniformly distributed over $(0, 5)$. That is, Λ is uniformly distributed over $(0, 5)$, and given that $\Lambda = \lambda$, the number of storms is Poisson with mean λ. Find the probability there are at least three storms this season.

Exercise 58. A collection of n coins is flipped. The outcomes are independent, and the ith coin comes up heads with probability $\alpha_i, i = 1, \ldots, n$. Suppose that for some value of $j, 1 \leqslant j \leqslant n, \alpha_j = \frac{1}{2}$. Find the probability that the total number of heads to appear on the n coins is an even number.

Exercise 59. Suppose each new coupon collected is, independent of the past, a type i coupon with probability p_i. A total of n coupons is to be collected. Let A_i be the event that there is at least one type i in this set. For $i \neq j$, compute $P(A_iA_j)$ by (a) conditioning on N_i, the number of type i coupons in the set of n coupons; (b) conditioning on F_i, the first time a type i coupon is collected; (c) using the identity $P(A_i \cup A_j) = P(A_i) + P(A_j) - P(A_iA_j)$.

Exercise 60. *Two players alternate flipping a coin that comes up heads with probability p. The first player to obtain a head is declared the winner. We are interested in the probability that the first player to flip is the winner. Before determining this probability, which we will call $f(p)$, answer the following questions. (a) Do you think that $f(p)$ is a monotone function of p? If so, is it increasing or decreasing? (b) What do you think is the value of $\lim_{p\to 1} f(p)$? (c) What do you think is the value of $\lim_{p\to 0} f(p)$? (d) Find $f(p)$.

Exercise 61. Suppose in Exercise 29 that the shooting ends when the target has been hit twice. Let m_i denote the mean number of shots needed for the first hit when player i shoots first,

$i = 1, 2$. Also, let $P_i, i = 1, 2$, denote the probability that the first hit is by player 1, when player i shoots first. (a) Find m_1 and m_2. (b) Find P_1 and P_2. For the remainder of the problem, assume that player 1 shoots first. (c) Find the probability that the final hit was by 1. (d) Find the probability that both hits were by 1. (e) Find the probability that both hits were by 2. (f) Find the mean number of shots taken.

Exercise 62. A, B, and C are evenly matched tennis players. Initially A and B play a set, and the winner then plays C. This continues, with the winner always playing the waiting player, until one of the players has won two sets in a row. That player is then declared the overall winner. Find the probability that A is the overall winner.

Exercise 63. Suppose there are n types of coupons, and that the type of each new coupon obtained is independent of past selections and is equally likely to be any of the n types. Suppose one continues collecting until a complete set of at least one of each type is obtained. (a) Find the probability that there is exactly one type i coupon in the final collection. Hint: Condition on T, the number of types that are collected before the first type i appears. (b) Find the expected number of types that appear exactly once in the final collection.

Exercise 64. A and B roll a pair of dice in turn, with A rolling first. A's objective is to obtain a sum of 6, and B's is to obtain a sum of 7. The game ends when either player reaches his or her objective, and that player is declared the winner. (a) Find the probability that A is the winner. (b) Find the expected number of rolls of the dice. (c) Find the variance of the number of rolls of the dice.

Exercise 65. The number of red balls in an urn that contains n balls is a random variable that is equally likely to be any of the values $0, 1, \ldots, n$. That is,

$$P\{i \text{ red}, n - i \text{ non-red}\} = \frac{1}{n+1}, \quad i = 0, 1, \ldots, n$$

The n balls are then randomly removed one at a time. Let Y_k denote the number of red balls in the first k selections, $k = 1, \ldots, n$. (a) Find $P\{Y_n = j\}, j = 0, \ldots, n$. (b) Find $P\{Y_{n-1} = j\}, j = 0, \ldots, n-1$. (c) What do you think is the value of $P\{Y_k = j\}, j = 0, \ldots, k$? (d) Verify your answer to part (c) by a backwards induction argument. That is, check that your answer is correct when $k = n$, and then show that whenever it is true for k it is also true for $k - 1, k = 1, \ldots, n$.

Exercise 66. The opponents of soccer team A are of two types: either they are a class 1 or a class 2 team. The number of goals team A scores against a class i opponent is a Poisson random variable with mean λ_i, where $\lambda_1 = 2, \lambda_2 = 3$. This weekend the team has two games against teams they are not very familiar with. Assuming that the first team they play is a class 1 team with probability 0.6 and the second is, independently of the class of the first team, a class 1 team with probability 0.3, determine (a) the expected number of goals team A will score this weekend. (b) the probability that team A will score a total of five goals.

Exercise 67. *A coin having probability p of coming up heads is continually flipped. Let $P_j(n)$ denote the probability that a run of j successive heads occurs within the first n flips. (a) Argue that

$$P_j(n) = P_j(n-1) + p^j(1-p)[1 - P_j(n-j-1)]$$

(b) By conditioning on the first non-head to appear, derive another equation relating $P_j(n)$ to the quantities $P_j(n-k), k = 1, \ldots, j$.

Exercise 68. In a knockout tennis tournament of 2^n contestants, the players are paired and play a match. The losers depart, the remaining 2^{n-1} players are paired, and they play a match.

This continues for n rounds, after which a single player remains unbeaten and is declared the winner. Suppose that the contestants are numbered 1 through 2^n, and that whenever two players contest a match, the lower numbered one wins with probability p. Also suppose that the pairings of the remaining players are always done at random so that all possible pairings for that round are equally likely. (a) What is the probability that player 1 wins the tournament? (b) What is the probability that player 2 wins the tournament?

Exercise 69. In the match problem, say that $(i, j), i < j$, is a pair if i chooses j's hat and j chooses i's hat. (a) Find the expected number of pairs. (b) Let Q_n denote the probability that there are no pairs, and derive a recursive formula for Q_n in terms of $Q_j, j < n$. (c) Use the recursion of part (b) to find Q_8.

Exercise 70. Let N denote the number of cycles that result in the match problem. (a) Let $M_n = E[N]$, and derive an equation for M_n in terms of $M_1, \ldots, M_{n-1}$. (b) Let C_j denote the size of the cycle that contains person j. Argue that

$$N = \sum_{j=1}^{n} 1/C_j$$

and use the preceding to determine $E[N]$. (c) Find the probability that persons $1, 2, \ldots, k$ are all in the same cycle. (d) Find the probability that $1, 2, \ldots, k$ is a cycle.

Exercise 71. Let $M_0 = 0$ and suppose that, for $n \geq 1$,

$$M_n = 1 + \frac{1}{n} \sum_{i=0}^{n-1} M_i.$$

By subtracting suitable equations for n and $n - 1$, show that $M_n = \sum_{i=1}^{n} 1/i$.

Exercise 72. Let $U_1, U_2, \ldots$ be independent uniform random variables on $(0, 1)$, and define

$$N = \min\{n \geq 2 : U_n > U_{n-1}\}, \qquad M = \min\{n \geq 1 : U_1 + \cdots + U_n > 1\}.$$

For $0 < y < 1$, show that the conditional distribution of N given $U_1 = y$ equals that of M given $U_1 = 1 - y$, and show that both conditional expectations equal $1 + e^y$.

Exercise 76. Consider a gambler who on each bet either wins 1 with probability 18/38 or loses 1 with probability 20/38. (These are the probabilities if the bet is that a roulette wheel will land on a specified color.) The gambler will quit either when he or she is winning a total of 5 or after 100 plays. What is the probability he or she plays exactly 15 times?

Exercise 77. Show that (a) $E[XY \mid Y = y] = yE[X \mid Y = y]$ (b) $E[g(X, Y) \mid Y = y] = E[g(X, y) \mid Y = y]$ (c) $E[XY] = E[YE[X \mid Y]]$

Exercise 79. An urn contains n white and m black balls that are removed one at a time. If $n > m$, show that the probability that there are always more white than black balls in the urn (until, of course, the urn is empty) equals $(n - m)/(n + m)$. Explain why this probability is equal to the probability that the set of withdrawn balls always contains more white than black balls. (This latter probability is $(n - m)/(n + m)$ by the ballot problem.)

Exercise 82. Let $X_1, X_2, \ldots$ be independent continuous random variables with a common distribution function F and density $f = F'$, and for $k \geq 1$ let

$$N_k = \min\{n \geq k : X_n = k\text{th largest of } X_1, \ldots, X_n\}$$

(a) Show that $P\{N_k = n\} = \frac{k-1}{n(n-1)}$, $n \geq k$.
(b) Argue that

$$f_{X_{N_k}}(x) = f(x)(\bar{F}(x))^{k-1} \sum_{i=0}^{\infty} \binom{i+k-2}{i} (F(x))^i$$

(c) Prove the following identity:

$$a^{1-k} = \sum_{i=0}^{\infty} \binom{i+k-2}{i} (1-a)^i, \quad 0 < a < 1,\ k \geq 2$$

Hint: Use induction. First prove it when $k = 2$, and then assume it for k. To prove it for $k+1$, use the fact that

$$\sum_{i=1}^{\infty} \binom{i+k-1}{i} (1-a)^i = \sum_{i=1}^{\infty} \binom{i+k-2}{i} (1-a)^i + \sum_{i=1}^{\infty} \binom{i+k-2}{i-1} (1-a)^i$$

where the preceding used the combinatorial identity $\binom{m}{i} = \binom{m-1}{i} + \binom{m-1}{i-1}$. Now, use the induction hypothesis to evaluate the first term on the right side of the preceding equation.
(d) Conclude that X_{N_k} has distribution F.

Exercise 83. An urn contains n balls, with ball i having weight $w_i, i = 1, \ldots, n$. The balls are withdrawn from the urn one at a time according to the following scheme: When S is the set of balls that remains, ball $i, i \in S$, is the next ball withdrawn with probability $w_i / \sum_{j \in S} w_j$. Find the expected number of balls that are withdrawn before ball $i, i = 1, \ldots, n$.

Exercise 86. On five labelled vertices, each vertex independently chooses one of the five vertices uniformly as its directed-arc destination; self-loops are allowed. A component means a weakly connected component of this random mapping. Compute the probability distribution of the number of components and verify your solution by using it to compute $E[C]$ and then comparing your solution with

$$E[C] = \sum_{k=1}^{5} \binom{5}{k} \frac{(k-1)!}{5^k}$$

Exercise 91. Find the expected number of flips of a coin, which comes up heads with probability p, that are necessary to obtain the pattern h, t, h, h, t, h, t, h.

Exercise 93. Consider a sequence of independent trials, each of which is equally likely to result in any of the outcomes $0, 1, \ldots, m$. Say that a round begins with the first trial, and that a new round begins each time outcome 0 occurs. Let N denote the number of trials that it takes until all of the outcomes $1, \ldots, m-1$ have occurred in the same round. Assume $m \geq 2$. Outcome m is neutral. Let T_j denote the number of trials until j distinct outcomes from $\{0, \ldots, m-1\}$ have occurred, and let I_j be the jth distinct outcome from that relevant set. Neutral trials count toward elapsed time, but not toward discoveries. (a) Argue that the random vectors $(I_1, \ldots, I_m)$ and $(T_1, \ldots, T_m)$ are independent.
(b) Define X by letting $X = j$ if outcome 0 is the jth distinct outcome to occur. Derive an equation for $E[N]$ in terms of $E[T_j], j = 1, \ldots, m-1$ by conditioning on X.
(c) Determine $E[T_j], j = 1, \ldots, m-1$.
(d) Find $E[N]$.

Exercise 94. Let N be a hypergeometric random variable having the distribution of the number of white balls in a random sample of size r from a set of w white and b blue balls. That is,

$$P\{N = n\} = \frac{\binom{w}{n}\binom{b}{r-n}}{\binom{w+b}{r}}$$

where we use the convention $\binom{m}{j} = 0$ if either $j < 0$ *or* $j > m$.

Assume $w \geq 1$, $b \geq 0$, and $1 \leq r \leq w + b$. Consider $S_N = \sum_{i=1}^{N} X_i$, where the X_i are i.i.d. positive integer-valued random variables, independent of N, with $\alpha_j = P\{X_i = j\}$.
(a) Define M by the size-biased law $P(M = n) = nP(N = n)/E[N]$. Find the distribution of $M - 1$.
(b) Suppressing its dependence on b, let $P_{w,r}(k) = P\{S_N = k\}$, and derive a recursion equation for $P_{w,r}(k)$.
(c) Use the recursion of (b) to find $P_{w,r}(2)$.

Exercise 95. Let $X_1, X_2, \ldots$ be i.i.d. integer-valued steps with $X_i \geq -1$ and $-\infty < E[X_i] < 0$, and let $S_0 = 0$, $S_n = \sum_{i=1}^{n} X_i$. Define $\beta = P\{S_n \leq 0 \text{ for all } n\}$ be the probability that the walk is never positive. Find β when $E[X_i] < 0$.

Exercise 96. Consider a large population of families, and suppose that the number of children in the different families are independent Poisson random variables with mean λ. Show that the number of siblings of a randomly chosen child is also Poisson distributed with mean λ.

4 Distribution Recursions

Exercise 89. Let $I_1, \ldots, I_n$ be independent random variables, each of which is equally likely to be either 0 or 1. A well-known nonparametric statistical test, called the signed rank test, is concerned with determining $P_n(k)$ defined by

$$P_n(k) = P\left\{\sum_{j=1}^{n} jI_j \leq k\right\}.$$

Justify the following formula:

$$P_n(k) = \frac{1}{2}P_{n-1}(k) + \frac{1}{2}P_{n-1}(k-n).$$

5 Monte Carlo and Simulation Methods

Exercise 5. Suppose it is relatively easy to simulate from F_i for each $i = 1, \ldots, n$. How can we simulate from

(a) $F(x) = \prod_{i=1}^{n} F_i(x)$?

(b) $F(x) = 1 - \prod_{i=1}^{n}(1 - F_i(x))$?

(c) Give two methods for simulating from the distribution $F(x) = x^n$, $0 < x < 1$.

Exercise 6. Consider simulating the absolute value of a standard normal random variable using rejection sampling with an exponential proposal. For a positive rate λ, the proposal density is

$$g(x) = \lambda e^{-\lambda x}.$$

Show that the mean number of iterations needed in the rejection scheme is minimized when $\lambda = 1$.

Exercise 7. Give an algorithm for simulating a random variable having density function

$$f(x) = 30(x^2 - 2x^3 + x^4), \qquad 0 < x < 1.$$

Exercise 8. Consider the technique of simulating a gamma (n, λ) random variable by using the rejection method with g being an exponential density with rate λ/n.

(a) Show that the average number of iterations of the algorithm needed to generate a gamma is

$$\frac{n^n e^{1-n}}{(n-1)!}.$$

(b) Use Stirling's approximation to show that for large n the answer to part (a) is approximately equal to

$$e\left[\frac{n-1}{2\pi}\right]^{1/2}.$$

(c) Show that the procedure is equivalent to the following:

Step 1: Generate Y_1 and Y_2 independent exponentials with rate 1.

Step 2: If $Y_1 < (n-1)[Y_2 - \log(Y_2) - 1]$, return to step 1.

Step 3: Set $X = \frac{nY_2}{\lambda}$.

(d) Explain how to obtain an independent exponential along with a gamma from the preceding algorithm.

Exercise 13. The Discrete Rejection Method. Suppose we want to simulate X having probability mass function

$$P\{X = i\} = P_i, \qquad i = 1, \ldots, n,$$

and suppose we can easily simulate from the probability mass function Q_i, where

$$\sum_i Q_i = 1, \qquad Q_i \geq 0.$$

Let C be such that

$$P_i \leq CQ_i, \qquad i = 1, \ldots, n.$$

Show that the following algorithm generates the desired random variable:

Step 1: Generate Y having mass function Q and U an independent random number.

Step 2: If $U \leq \dfrac{P_Y}{CQ_Y}$, set $X = Y$. Otherwise return to step 1.

Exercise 16. Suppose n balls having weights $w_1, w_2, \ldots, w_n$ are in an urn. These balls are sequentially removed in the following manner: At each selection, a given ball in the urn is chosen with probability equal to its weight divided by the sum of the weights of all balls still in the urn (including the chosen ball). Let $I_1, I_2, \ldots, I_n$ denote the order in which the balls are removed, so that $I_1, \ldots, I_n$ is a random permutation with weights.

(a) Give a method for simulating $I_1, \ldots, I_n$.

(b) Let X_i be independent exponentials with rates w_i, $i = 1, \ldots, n$. Explain how X_i can be utilized to simulate $I_1, \ldots, I_n$.

Exercise 23. For a nonhomogeneous Poisson process with intensity function $\lambda(t)$, $t \geq 0$, where

$$\int_0^\infty \lambda(t)\, dt = \infty,$$

let $X_1, X_2, \ldots$ denote the sequence of times at which events occur.

(a) Show that

$$\int_0^{X_1} \lambda(t)\, dt$$

is exponential with rate 1.

(b) Show that

$$\int_{X_{i-1}}^{X_i} \lambda(t)\, dt, \qquad i \geq 1,$$

are independent exponentials with rate 1, where $X_0 = 0$.

In words, independent of the past, the additional amount of hazard that must be experienced until an event occurs is exponential with rate 1.

Exercise 25. Let (X, Y) be uniformly distributed in a circle of radius r about the origin. That is, their joint density is given by

$$f(x, y) = \frac{1}{\pi r^2}, \qquad 0 \leq x^2 + y^2 \leq r^2.$$

Let

$$R = \sqrt{X^2 + Y^2} \qquad \text{and} \qquad \theta = \operatorname{atan2}(Y, X) \pmod{2\pi}$$

denote their polar coordinates. Show that R and θ are independent with θ being uniform on $(0, 2\pi)$ and

$$P\{R < a\} = \frac{a^2}{r^2}, \qquad 0 < a < r.$$

Exercise 31. Consider a queueing system in which each service time, independent of the past, has mean μ. Let W_n and D_n denote, respectively, the amounts of time customer n spends in the system and in queue. Hence,

$$D_n = W_n - S_n,$$

where S_n is the service time of customer n. Therefore,

$$E[D_n] = E[W_n] - \mu.$$

If we use simulation to estimate $E[D_n]$, should we

(a) use the simulated data to determine D_n, which is then used as an estimate of $E[D_n]$, or

(b) use the simulated data to determine W_n and then use this quantity minus μ as an estimate of $E[D_n]$?

Repeat for when we want to estimate $E[W_n]$.

Exercise 32. Show that if X and Y have the same distribution, then

$$\mathrm{Var}\left(\frac{X+Y}{2}\right) \leq \mathrm{Var}(X).$$

Hence, conclude that the use of antithetic variables can never increase variance, though it need not be as efficient as generating an independent set of random numbers.

Exercise 33. If $0 \leq X \leq a$, show that

(a) $E[X^2] \leq aE[X]$,

(b) $\mathrm{Var}(X) \leq E[X](a - E[X])$,

(c) $\mathrm{Var}(X) \leq a^2/4$.

Exercise 35. Suppose we are able to simulate independent random variables X and Y. If we simulate $2k$ independent random variables

$$X_1, \ldots, X_k$$

and

$$Y_1, \ldots, Y_k,$$

where the X_i have the same distribution as does X and the Y_j have the same distribution as does Y, how would you use them to estimate

$$P(X < Y)?$$

6 Markov Chains

Notation and Setup Throughout, $P = (P_{ij})$ denotes a transition probability matrix, $\pi = (\pi_j)$ the stationary distribution, and $P_{ij}^{(n)}$ the n-step transition probability. The following models and state orderings are used in this chapter:

- **Gary's mood:** States $0 = C$ (cheerful), $1 = S$ (so-so), $2 = G$ (glum).
$$P = \begin{pmatrix} 0.5 & 0.4 & 0.1 \\ 0.3 & 0.4 & 0.3 \\ 0.2 & 0.3 & 0.5 \end{pmatrix}.$$
- **Rain depends on the last two days:** States $0 = (RR)$, $1 = (RN)$, $2 = (NR)$, $3 = (NN)$ (R = rain, N = no rain, ordered today/yesterday).
$$P = \begin{pmatrix} 0.7 & 0 & 0.3 & 0 \\ 0.5 & 0 & 0.5 & 0 \\ 0 & 0.4 & 0 & 0.6 \\ 0 & 0.2 & 0 & 0.8 \end{pmatrix}.$$
- **Two-ball urn in Exercise 9:** State = # red balls.
$$P = \begin{pmatrix} 0.8 & 0.2 & 0 \\ 0.1 & 0.8 & 0.1 \\ 0 & 0.2 & 0.8 \end{pmatrix}.$$
- **Random walk:** $P_{i,i+1} = p = 1 - P_{i,i-1}$, $i \in \mathbb{Z}$.

Exercise 1. Three white and three black balls are distributed in two urns so that each contains three balls. The system is in state i, $i = 0, 1, 2, 3$, if the first urn contains i white balls. At each step, one ball is drawn from each urn and the two balls are swapped. Let X_n be the state after the n-th step. Explain why $\{X_n\}$ is a Markov chain and calculate its transition probability matrix.

Exercise 2. Suppose that whether or not it rains today depends on previous weather conditions through the *last three days*. Show how this system may be analysed by a Markov chain. How many states are needed?

Exercise 3. In Exercise 2, suppose: if it has rained for the past three days, it will rain today with probability 0.8; if it did not rain for any of the past three days, it will rain today with probability 0.2; in any other case, the weather today will with probability 0.6 be the same as yesterday. Determine P for this Markov chain.

Exercise 4. Consider $\{X_n, n \geq 0\}$ taking values $0, 1, 2$, with
$$P\{X_{n+1} = j \mid X_n = i, \ldots\} = \begin{cases} P_{ij}^{\mathrm{I}}, & n \text{ even}, \\ P_{ij}^{\mathrm{II}}, & n \text{ odd}, \end{cases}$$
where $\sum_j P_{ij}^{\mathrm{I}} = \sum_j P_{ij}^{\mathrm{II}} = 1$. Is $\{X_n\}$ a Markov chain? If not, show how to transform it into one by enlarging the state space.

Exercise 5. A Markov chain $\{X_n, n \geq 0\}$ with states $0, 1, 2$ has transition matrix
$$P = \begin{pmatrix} \frac{1}{2} & \frac{1}{3} & \frac{1}{6} \\ 0 & \frac{1}{3} & \frac{2}{3} \\ \frac{1}{2} & 0 & \frac{1}{2} \end{pmatrix}.$$

If $P\{X_0 = 0\} = P\{X_0 = 1\} = \frac{1}{4}$, find $\mathbb{E}[X_3]$.

Exercise 6. Let the transition matrix of a two-state Markov chain be

$$\mathbf{P} = \begin{Vmatrix} p & 1-p \\ 1-p & p \end{Vmatrix}.$$

Show by induction that

$$\mathbf{P}^{(n)} = \begin{Vmatrix} \frac{1}{2} + \frac{1}{2}(2p-1)^n & \frac{1}{2} - \frac{1}{2}(2p-1)^n \\ \frac{1}{2} - \frac{1}{2}(2p-1)^n & \frac{1}{2} + \frac{1}{2}(2p-1)^n \end{Vmatrix}.$$

Exercise 7. In the rain model given in the chapter setup, suppose it has rained neither yesterday nor the day before yesterday. What is the probability that it will rain tomorrow?

Exercise 8. Coin 1 has $P(H) = 0.7$ and coin 2 has $P(H) = 0.6$. If today's flip is heads we use coin 1 tomorrow; if tails we use coin 2 tomorrow. The initial coin is equally likely to be coin 1 or coin 2.

(a) What is the probability that the coin flipped on the third day after the initial flip is coin 1?

(b) Suppose the coin flipped on Monday comes up heads. What is the probability that the coin flipped on Friday of the same week also comes up heads?

Exercise 9. An urn initially contains one red and one blue ball. At each step, select one of its two balls uniformly, record its colour, and replace it. With probability 0.8 the replacement has the same colour; with probability 0.2 replace it by a ball of the opposite colour. These random choices are independent between steps. Let $X_n = 1$ if the nth selected ball is red, and $X_n = 0$ otherwise. Is $\{X_n : n \geq 1\}$ a Markov chain?

Exercise 10. In the mood model given in the chapter setup, Gary is currently in a cheerful mood. What is the probability that he is *not* in a glum mood on any of the following three days?

Exercise 11. In the mood model given in the chapter setup, Gary was in a glum mood four days ago. Given that he hasn't felt cheerful in a week, what is the probability he is feeling glum today?

Exercise 14. Specify the classes of the following Markov chains, and determine whether they are transient or recurrent:

$$P_1 = \begin{Vmatrix} 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 0 \end{Vmatrix}, \quad P_2 = \begin{Vmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 0 & 0 & 1 & 0 \end{Vmatrix},$$

$$P_3 = \begin{Vmatrix} \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} & 0 & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 0 & \frac{1}{2} & \frac{1}{2} \end{Vmatrix}, \quad P_4 = \begin{Vmatrix} \frac{1}{4} & \frac{3}{4} & 0 & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & \frac{1}{3} & \frac{2}{3} & 0 \\ 1 & 0 & 0 & 0 & 0 \end{Vmatrix}.$$

Exercise 16. Show that if state i is recurrent and state i does not communicate with state j, then $P_{ij} = 0$.

Exercise 18. Coin 1 comes up heads with probability 0.6 and coin 2 with probability 0.5. A coin is continually flipped until it comes up tails, at which time that coin is put aside and we start flipping the other one.

(a) What proportion of flips use coin 1?

(b) If we start with coin 1, what is the probability that coin 2 is used on the fifth flip?

Exercise 21. A DNA nucleotide has any of four values. In the standard mutation model, from period to period a nucleotide at a specific location does not change with probability $1-3\alpha$, and if it changes it is equally likely to become any of the other three values ($0 < \alpha < \frac{1}{3}$).

(a) Show that $P^n_{1,1} = \frac{1}{4} + \frac{3}{4}(1-4\alpha)^n$.

(b) What is the long-run proportion of time the chain is in each state?

Exercise 23. In a good year, storms are Poisson with mean 1; in a bad year, Poisson with mean 3. A good year is equally likely to be followed by a good or bad year; a bad year is twice as likely to be followed by a bad year as by a good year. Last year (year 0) was good.

(a) Find the expected total number of storms in years 1 and 2.

(b) Find the probability of no storms in year 3.

(c) Find the long-run average storms per year.

Exercise 24. Consider three urns: red (1 red, 4 blue), white (3 white, 2 red, 2 blue), blue (4 white, 3 red, 2 blue). At each stage a ball is selected from the urn whose colour matches the previously selected ball. In the long run, what proportion of the selected balls are red / white / blue?

Exercise 25. An individual owns k pairs of running shoes. Each morning he leaves by either door (front or back) with equal probability 1/2, taking shoes if available; on return he enters by either door with equal probability 1/2, leaving shoes there. What proportion of the time does he run barefooted?

Exercise 26. Consider the following approach to shuffling a deck of n cards: choose one of the positions $1, 2, \ldots, n$ uniformly at random and move the card in that position to the top of the deck ("move to front"). Show that in the limit the deck is perfectly shuffled, i.e. every ordering is equally likely.

Exercise 27. Each of N individuals is either active or inactive each period. An active individual stays active w.p. α; an inactive individual stays inactive w.p. β. Let X_n = number of active individuals in period n.

(a) Argue that X_n is a Markov chain.

(b) Find $\mathbb{E}[X_n \mid X_0 = i]$.

(c) Derive an expression for the transition probabilities.

(d) Find the long-run proportion of time that exactly j individuals are active.

Exercise 28. Every time the team wins a game it wins the next with probability 0.8; every time it loses it wins the next with probability 0.3. If it wins it has dinner with probability 0.7; if it loses, with probability 0.2. What proportion of games result in a team dinner?

Exercise 29. An organisation has N employees with three job classifications, changing in-

dependently according to $P = \begin{pmatrix} 0.7 & 0.2 & 0.1 \\ 0.2 & 0.6 & 0.2 \\ 0.1 & 0.4 & 0.5 \end{pmatrix}$. What percentage of employees are in each classification?

Exercise 30. Three out of every four trucks are followed by a car; one out of every five cars is followed by a truck. What fraction of vehicles on the road are trucks?

Exercise 31. A town never has two sunny days in a row. Days are sunny (S), cloudy (C), or rainy (R). If sunny, next day is equally likely C or R. If rainy or cloudy, probability $1/2$ of staying the same; if it changes, it is equally likely to be either of the other two. What proportion of days are sunny / cloudy?

Exercise 32. Each of two switches is either on or off each day. On day n, each switch independently is on with probability $[1 + (\text{number on during day } n-1)]/4$. What fraction of days are both switches on? Both off?

Exercise 33. A professor gives three types of exams. p_i = probability of doing well on type i ($p_1 = 0.3$, $p_2 = 0.6$, $p_3 = 0.9$). If the class does well, the next exam is equally likely to be any type; if the class does badly, the next exam is always type 1. What proportion of exams are of type i, $i = 1, 2, 3$?

Exercise 34. A flea moves on a triangle with vertices $1, 2, 3$. At vertex i it moves clockwise (CW) w.p. p_i and counterclockwise (CCW) w.p. $q_i = 1 - p_i$.

(a) Find the proportion of time at each vertex.

(b) How often does a CCW move immediately precede five consecutive CW moves?

Exercise 35. Consider a Markov chain with states $\{0, 1, 2, 3, 4\}$, where $P_{0,4} = 1$ and from state $i > 0$ the next state is equally likely to be any of $0, 1, \ldots, i-1$. Find the limiting probabilities.

Exercise 36. A process changes state daily according to the two-state Markov chain with $P_{0,0} = 0.4$, $P_{0,1} = 0.6$, $P_{1,0} = 0.2$, $P_{1,1} = 0.8$. Each day a message is sent: "good" w.p. p_i or "bad" w.p. $q_i = 1 - p_i$ if the chain is in state i.

(a) State 0 on Monday: probability of a good message on Tuesday?

(b) State 0 on Monday: probability of a good message on Friday?

(c) In the long run, what proportion of messages are good?

(d) Let $Y_n = 1$ if a good message is sent on day n, $Y_n = 2$ otherwise. Is $\{Y_n\}$ a Markov chain?

Exercise 38. Recall that state i is said to be positive recurrent if $m_{i,i} < \infty$, where $m_{i,i}$ is the expected number of transitions until the Markov chain, starting in state i, makes a transition back into that state. Because π_i, the long-run proportion of time the Markov chain, starting in state i, spends in state i, satisfies

$$\pi_i = \frac{1}{m_{i,i}}$$

it follows that state i is positive recurrent if and only if $\pi_i > 0$. Suppose that state i is positive recurrent and that state i communicates with state j. Show that state j is also positive recurrent by arguing that there is an integer n such that

$$\pi_j \geq \pi_i P_{i,j}^n > 0$$

Exercise 39. Recall that a recurrent state that is not positive recurrent is called null recurrent. Use the result of Exercise 38 to prove that null recurrence is a class property. That is, if state i is null recurrent and state i communicates with state j, show that state j is also null recurrent.

Exercise 40. It follows from the argument made in Exercise 38 that state i is null recurrent if it is recurrent and $\pi_i = 0$. Consider the symmetric random walk on $\mathbb{Z}$, with independent steps $+1$ and -1 of probability $1/2$ each.

(a) Argue that $\pi_i = \pi_0$ for all i.

(b) Argue that all states are null recurrent.

Exercise 41. Let π_i denote the long-run proportion of time a given irreducible Markov chain is in state i.

(a) Explain why π_i is also the proportion of transitions that are into state i as well as being the proportion of transitions that are from state i.

(b) $\pi_i P_{i,j}$ represents the proportion of transitions that satisfy what property?

(c) $\sum_i \pi_i P_{i,j}$ represent the proportion of transitions that satisfy what property?

(d) Using the preceding explain why
$$\pi_j = \sum_i \pi_i P_{i,j}$$

Exercise 42. Let A be a set of states, and let A^c be the remaining states.

(a) What is the interpretation of
$$\sum_{i \in A} \sum_{j \in A^c} \pi_i P_{i,j}?$$

(b) What is the interpretation of
$$\sum_{i \in A^c} \sum_{j \in A} \pi_i P_{i,j}?$$

(c) Explain the identity
$$\sum_{i \in A} \sum_{j \in A^c} \pi_i P_{i,j} = \sum_{i \in A^c} \sum_{j \in A} \pi_i P_{i,j}$$

Exercise 43. Each day, one of n possible elements is requested, the ith one with probability $P_i, i \geq 1, \sum_1^n P_i = 1$. These elements are at all times arranged in an ordered list that is revised as follows: The element selected is moved to the front of the list with the relative positions of all the other elements remaining unchanged. Define the state at any time to be the list ordering at that time and note that there are $n!$ possible states.

(a) Argue that the preceding is a Markov chain.

(b) For any state $i_1, \ldots, i_n$ (which is a permutation of $1, 2, \ldots, n$), let $\pi(i_1, \ldots, i_n)$ denote the limiting probability. In order for the state to be $i_1, \ldots, i_n$, it is necessary for the last request to be for i_1, the last non-i_1 request for i_2, the last non-i_1 or i_2 request for i_3, and so on. Hence, it appears intuitive that
$$\pi(i_1, \ldots, i_n) = P_{i_1} \frac{P_{i_2}}{1 - P_{i_1}} \frac{P_{i_3}}{1 - P_{i_1} - P_{i_2}} \cdots \frac{P_{i_{n-1}}}{1 - P_{i_1} - \cdots - P_{i_{n-2}}}$$
Verify when $n = 3$ that the preceding are indeed the limiting probabilities.

Exercise 44. Suppose that a population consists of a fixed number, say, m, of genes in any generation. Each gene is one of two possible genetic types. If exactly i (of the m) genes of any generation are of type 1, then the next generation will have j type 1 genes (and $m-j$ type 2 genes) with probability

$$\binom{m}{j}\left(\frac{i}{m}\right)^j\left(\frac{m-i}{m}\right)^{m-j}, \quad j=0,1,\ldots,m$$

Let X_n denote the number of type 1 genes in the nth generation, and assume that $X_0=i$.

(a) Find $E[X_n]$.

(b) What is the probability that eventually all the genes will be type 1?

Exercise 46. An individual possesses r umbrellas that he employs in going from his home to office, and vice versa. If he is at home (the office) at the beginning (end) of a day and it is raining, then he will take an umbrella with him to the office (home), provided there is one to be taken. If it is not raining, then he never takes an umbrella. Assume that, independent of the past, it rains at the beginning (end) of a day with probability p, independent of the past.

(a) Define a Markov chain with $r+1$ states, which will help us to determine the proportion of time that our man gets wet. (Note: He gets wet if it is raining, and all umbrellas are at his other location.)

(b) Show that the limiting probabilities are given by

$$\pi_i=\begin{cases}\dfrac{q}{r+q}, & \text{if } i=0\\[2ex] \dfrac{1}{r+q}, & \text{if } i=1,\ldots,r\end{cases} \qquad \text{where } q=1-p$$

(c) What fraction of time does our man get wet?

(d) When $r=3$, what value of p maximizes the fraction of time he gets wet?

Exercise 47. Let $\{X_n, n\geq 0\}$ denote an ergodic Markov chain with limiting probabilities π_i. Define the process $\{Y_n, n\geq 1\}$ by $Y_n=(X_{n-1},X_n)$. That is, Y_n keeps track of the last two states of the original chain. Is $\{Y_n, n\geq 1\}$ a Markov chain? If so, determine its transition probabilities and find $\lim_{n\to\infty} P\{Y_n=(i,j)\}$.

Exercise 48. Consider a Markov chain in steady state. Say that a k length run of zeroes ends at time m if

$$X_{m-k-1}\neq 0, \quad X_{m-k}=X_{m-k+1}=\cdots=X_{m-1}=0, \quad X_m\neq 0$$

Show that the probability of this event is $\pi_0(P_{0,0})^{k-1}(1-P_{0,0})^2$, where π_0 is the limiting probability of state 0.

Exercise 49. Let $P^{(1)}$ and $P^{(2)}$ denote transition probability matrices for ergodic Markov chains having the same state space. Let π^1 and π^2 denote the stationary (limiting) probability vectors for the two chains. Consider a process defined as follows:

(a) $X_0=1$. A coin is then flipped and if it comes up heads, then the remaining states $X_1,\ldots$ are obtained from the transition probability matrix $P^{(1)}$ and if tails from the matrix $P^{(2)}$. Is $\{X_n, n\geq 0\}$ a Markov chain? If $p=P\{\text{coin comes up heads}\}$, what is $\lim_{n\to\infty} P(X_n=i)$?

(b) $X_0=1$. At each stage the coin is flipped and if it comes up heads, then the next state is chosen according to $P^{(1)}$ and if tails comes up, then it is chosen according to $P^{(2)}$. In this

case do the successive states constitute a Markov chain? If so, determine the transition probabilities. Show by a counterexample that the limiting probabilities are not the same as in part (a).

Exercise 50. In Exercise 8, if today's flip lands heads, what is the expected number of additional flips needed until the pattern t, t, h, t, h, t, t occurs?

Exercise 51. In the mood model given in the chapter setup, Gary is in a cheerful mood today. Find the expected number of days until he has been glum for three consecutive days.

Exercise 52. A taxi driver provides service in two zones of a city. Fares picked up in zone A will have destinations in zone A with probability 0.6 or in zone B with probability 0.4. Fares picked up in zone B will have destinations in zone A with probability 0.3 or in zone B with probability 0.7. The driver's expected profit for a trip entirely in zone A is 6; for a trip entirely in zone B is 8; and for a trip that involves both zones is 12. Find the taxi driver's average profit per trip.

Exercise 53. An insurer assigns each policyholder an annual premium level 0, 1, or 2, charging 500, 750, or 1000 monetary units, respectively. The level at the start of a year determines that year's premium. For a policyholder of type λ, annual claim counts are independent Poisson random variables with mean λ. A claim-free year lowers the next year's level by one, stopping at level 0; a year with one or more claims sets the next year's level to 2. A policyholder's type remains fixed. One-third of policyholders have $\lambda = 1/4$ and two-thirds have $\lambda = 1/2$. Find the long-run average annual premium per policyholder.

Exercise 54. Consider the Ehrenfest urn model in which M molecules are distributed between two urns, and at each time point one of the molecules is chosen at random and is then removed from its urn and placed in the other one. Let X_n denote the number of molecules in urn 1 after the nth switch and let $\mu_n = E[X_n]$. Show that

(a) $\mu_{n+1} = 1 + (1 - 2/M)\mu_n$.

(b) Use (a) to prove that

$$\mu_n = \frac{M}{2} + \left(\frac{M-2}{M}\right)^n \left(E[X_0] - \frac{M}{2}\right)$$

Exercise 55. Consider a population of individuals each of whom possesses two genes that can be either type A or type a, with A dominant. For randomly and independently chosen parents, let S_{10} be the probability of a recessive child given one dominant and one recessive parent, and S_{11} the probability given two dominant parents. Compute S_{11} and S_{10} to show that $S_{11} = S_{10}^2$. (The quantities S_{10} and S_{11} are known in the genetics literature as Snyder's ratios.)

Exercise 56. Suppose that on each play of the game a gambler either wins 1 with probability p or loses 1 with probability $1 - p$. The gambler continues betting until she or he is either up n or down m. What is the probability that the gambler quits a winner?

Exercise 57. A particle moves among $n + 1$ vertices on a circle, moving clockwise with probability p and counterclockwise with probability $1 - p$ at each independent step. Starting at a specified state, call it state 0, let T be the time of the first return to state 0. Find the probability that all states have been visited by time T. **Hint:** Condition on the initial transition and then use results from the gambler's ruin problem.

Exercise 58. In a gambler's ruin game with fortunes $0, 1, \ldots, N$, each independent bet increases the fortune by 1 with probability $p \in (0, 1)$ or decreases it by 1 with probability $q = 1 - p$; play

stops at 0 or N. Suppose the gambler's fortune is presently i, and suppose that we know that the gambler's fortune will eventually reach N (before it goes to 0). Given this information, show that the probability he wins the next gamble is

$$\begin{cases} \dfrac{p[1-(q/p)^{i+1}]}{1-(q/p)^i}, & \text{if } p \neq \frac{1}{2} \\ \dfrac{i+1}{2i}, & \text{if } p = \frac{1}{2} \end{cases}$$

Exercise 59. For independent bets that change the fortune by $+1$ with probability $p \in (0,1)$ and -1 with probability $q = 1-p$, stopping at 0 or N, let M_i denote the mean number of games that must be played until the gambler either goes broke or reaches a fortune of N, given that he starts with $i, i = 0, 1, \ldots, N$. Show that M_i satisfies

$$M_0 = M_N = 0; \quad M_i = 1 + pM_{i+1} + qM_{i-1}, \quad i = 1, \ldots, N-1$$

Exercise 60. Solve the equations given in Exercise 59 to obtain

$$M_i = \begin{cases} i(N-i), & \text{if } p = \frac{1}{2} \\ \dfrac{i}{q-p} - \dfrac{N}{q-p}\left(\dfrac{1-(q/p)^i}{1-(q/p)^N}\right), & \text{if } p \neq \frac{1}{2} \end{cases}$$

Exercise 61. Suppose in the gambler's ruin problem that the probability of winning a bet depends on the gambler's present fortune. Specifically, suppose that α_i is the probability that the gambler wins a bet when his or her fortune is i. Given that the gambler's initial fortune is i, let $P(i)$ denote the probability that the gambler's fortune reaches N before 0.

(a) Derive a formula that relates $P(i)$ to $P(i-1)$ and $P(i+1)$.

(b) Using the same approach as in the gambler's ruin problem, solve the equation of part (a) for $P(i)$.

(c) Suppose that i balls are initially in urn 1 and $N-i$ are in urn 2, and suppose that at each stage one of the N balls is randomly chosen, taken from whichever urn it is in, and placed in the other urn. Find the probability that the first urn becomes empty before the second.

Exercise 64. Consider a branching process having $\mu < 1$. Show that if $X_0 = 1$, then the expected number of individuals that ever exist in this population is given by $1/(1-\mu)$. What if $X_0 = n$?

Exercise 66. For a branching process, calculate π_0 when

(a) $P_0 = \frac{1}{4}, P_2 = \frac{3}{4}$

(b) $P_0 = \frac{1}{4}, P_1 = \frac{1}{2}, P_2 = \frac{1}{4}$

(c) $P_0 = \frac{1}{6}, P_1 = \frac{1}{2}, P_3 = \frac{1}{3}$

Exercise 68. (a) Show that the limiting probabilities of the reversed Markov chain are the same as for the forward chain by showing that they satisfy the equations

$$\pi_j = \sum_i \pi_i Q_{i,j}$$

(b) Give an intuitive explanation for the result of part (a).

Exercise 70. A total of m white and m black balls are distributed among two urns, with each urn containing m balls. At each stage, a ball is randomly selected from each urn and the two selected balls are interchanged. Let X_n denote the number of black balls in urn 1 after the nth interchange.

(a) Give the transition probabilities of the Markov chain $X_n, n \geq 0$.

(b) Without any computations, what do you think are the limiting probabilities of this chain?

(c) Find the limiting probabilities and show that the stationary chain is time reversible.

Exercise 71. It follows from Theorem 4.2 that for a time reversible Markov chain

$$P_{i,j}P_{j,k}P_{k,i} = P_{i,k}P_{k,j}P_{j,i}, \quad \text{for all } i, j, k$$

It turns out that if the state space is finite and $P_{i,j} > 0$ for all i, j, then the preceding is also a sufficient condition for time reversibility. (That is, in this case, we need only check the equation for paths from i to i that have only two intermediate states.) Prove this.
Hint: Fix i and show that the equations

$$\pi_j P_{j,k} = \pi_k P_{k,j}$$

are satisfied by $\pi_j = cP_{i,j}/P_{j,i}$, where c is chosen so that $\sum_j \pi_j = 1$.

Exercise 74. A group of n processors is arranged in an ordered list. When a job arrives, the first processor in line attempts it; if it is unsuccessful, then the next in line tries it; if it too is unsuccessful, then the next in line tries it, and so on. When the job is successfully processed or after all processors have been unsuccessful, the job leaves the system. At this point we are allowed to reorder the processors, and a new job appears. Suppose that we use the one-closer reordering rule, which moves the processor that was successful one closer to the front of the line by interchanging its position with the one in front of it. If all processors were unsuccessful (or if the processor in the first position was successful), then the ordering remains the same. Suppose that each time processor i attempts a job then, independently of anything else, it is successful with probability p_i.

(a) Define an appropriate Markov chain to analyze this model.

(b) Show that this Markov chain is time reversible.

(c) Find the long-run probabilities.

7 Counting Processes

Exercise 1. The time T required to repair a machine is an exponentially distributed random variable with mean $\frac{1}{2}$ hours.

(a) What is the probability that a repair time exceeds $\frac{1}{2}$ hour?

(b) What is the probability that a repair takes at least $12\frac{1}{2}$ hours given that its duration exceeds 12 hours?

Exercise 2. Suppose that you arrive at a single-teller bank to find five other customers in the bank, one being served and the other four waiting in line. You join the end of the line. If the service times are all exponential with rate μ, what is the expected amount of time you will spend in the bank?

Exercise 3. Let X be an exponential random variable. Without any computations, tell which one of the following is correct. Explain your answer.

(a) $E[X^2 \mid X > 1] = E[(X+1)^2]$,

(b) $E[X^2 \mid X > 1] = E[X^2] + 1$,

(c) $E[X^2 \mid X > 1] = (1 + E[X])^2$.

Exercise 4. Consider a post office with two clerks. Three people, A, B, and C, enter simultaneously. A and B go directly to the clerks, and C waits until either A or B leaves before he begins service. What is the probability that A is still in the post office after the other two have left when

(a) the service time for each clerk is exactly ten minutes?

(b) the service times are i with probability $\frac{1}{3}$, $i = 1, 2, 3$?

(c) the service times are exponential with mean $1/\mu$?

Exercise 5. The lifetime of a radio is exponentially distributed with a mean of ten years. If Jones buys a ten-year-old radio, what is the probability that it will be working after an additional ten years?

Exercise 6. Smith arrives to find two servers busy and no one else waiting. The servers work independently, with exponential service times of rate λ_i at server i. Smith enters service with the first server to become free. Show that, among Smith and the two customers already in service,

$$P\{\text{Smith is not last}\} = \left(\frac{\lambda_1}{\lambda_1 + \lambda_2}\right)^2 + \left(\frac{\lambda_2}{\lambda_1 + \lambda_2}\right)^2.$$

Exercise 7. If X_1 and X_2 are independent nonnegative continuous random variables, show that

$$P\{X_1 < X_2 \mid \min(X_1, X_2) = t\} = \frac{r_1(t)}{r_1(t) + r_2(t)},$$

where $r_i(t)$ is the failure rate function of X_i.

Exercise 8. Let X_i, $i = 1, \ldots, n$, be independent continuous random variables, with X_i having failure rate function $r_i(t)$. Let T be independent of this sequence, and suppose that

$$\sum_{i=1}^{n} P\{T = i\} = 1.$$

Show that the failure rate function $r(t)$ of X_T is given by

$$r(t) = \sum_{i=1}^{n} r_i(t) P\{T = i \mid X_T > t\}.$$

Exercise 9. Machine 1 is currently working. Machine 2 will be put in use at a time t from now. If the lifetime of machine i is exponential with rate λ_i, $i = 1, 2$, what is the probability that machine 1 is the first machine to fail?

Exercise 10. Let X and Y be independent exponential random variables with respective rates λ and μ. Let $M = \min(X, Y)$. Find

(a) $E[MX \mid M = X]$,

(b) $E[MX \mid M = Y]$,

(c) $\mathrm{Cov}(X, M)$.

Exercise 11. Let $X, Y_1, \ldots, Y_n$ be independent exponential random variables, X having rate λ and Y_i having rate μ. Let A_i be the event that the ith smallest of these $n + 1$ random variables is one of the Y_i. Find

$$p = P\{X > \max_i Y_i\}$$

by using the identity

$$p = P(A_1 \cdots A_n) = P(A_1) P(A_2 \mid A_1) \cdots P(A_n \mid A_1 \cdots A_{n-1}).$$

Verify your answer when $n = 2$ by conditioning on X to obtain p.

Exercise 12. If X_i, $i = 1, 2, 3$, are independent exponential random variables with rates λ_i, $i = 1, 2, 3$, find

(a) $P\{X_1 < X_2 < X_3\}$,

(b) $P\{X_1 < X_2 \mid \max(X_1, X_2, X_3) = X_3\}$,

(c) $E[\max X_i \mid X_1 < X_2 < X_3]$,

(d) $E[\max X_i]$.

Exercise 13. A single server is busy and there are $n \geq 1$ customers waiting in first-come, first-served order. Service times are independent exponentials with rate μ. Independently, each waiting customer abandons the queue after an exponential patience time with rate $\theta > 0$. Find the expected time until the customer initially in position n in the queue leaves the queue, either by starting service or by abandoning it.

Exercise 14. Let X be an exponential random variable with rate λ.

(a) Use the definition of conditional expectation to determine $E[X \mid X < c]$.

(b) Now determine $E[X \mid X < c]$ by using the following identity:

$$E[X] = E[X \mid X < c] P(X < c) + E[X \mid X > c] P(X > c).$$

Exercise 15. One hundred items are simultaneously put on a life test. Suppose the lifetimes of the individual items are independent exponential random variables with mean 200 hours. The test will end when there have been a total of 5 failures. If T is the time at which the test ends, find $E[T]$ and $\mathrm{Var}(T)$.

Exercise 16. There are three jobs that need to be processed, with the processing time of job i being exponential with rate μ_i. There are two processors available, so processing on two of the jobs can immediately start, with processing on the final job to start when one of the initial ones is finished.

(a) Let T_i denote the time at which the processing of job i is completed. If the objective is to minimize $E[T_1 + T_2 + T_3]$, which jobs should be initially processed if $\mu_1 < \mu_2 < \mu_3$?

(b) Let M, called the makespan, be the time until all three jobs have been processed. With S equal to the time that there is only a single processor working, show that

$$2E[M] = E[S] + \sum_{i=1}^{3} \frac{1}{\mu_i}.$$

For the rest of this problem, suppose that $\mu_1 = \mu_2 = \mu$, $\mu_3 = \lambda$. Also, let $P(\mu)$ be the probability that the last job to finish is either job 1 or job 2, and let $P(\lambda) = 1 - P(\mu)$ be the probability that the last job to finish is job 3.

(c) Express $E[S]$ in terms of $P(\mu)$ and $P(\lambda)$.

(d) Let $P_{i,j}(\mu)$ be the value of $P(\mu)$ when i and j are the jobs that are initially started. Show that $P_{1,2}(\mu) \leq P_{1,3}(\mu)$.

(e) If $\mu > \lambda$, show that $E[M]$ is minimized when job 3 is one of the jobs that is initially started.

(f) If $\mu < \lambda$, show that $E[M]$ is minimized when processing is initially started on jobs 1 and 2.

Exercise 20. Consider a two-server system in which a customer is served first by server 1, then by server 2, and then departs. The service times at server i are exponential random variables with rates μ_i, $i = 1, 2$. When you arrive, you find server 1 free and two customers at server 2—customer A in service and customer B waiting in line.

(a) Find P_A, the probability that A is still in service when you move over to server 2.

(b) Find P_B, the probability that B is still in the system when you move over to server 2.

(c) Find $E[T]$, where T is the time that you spend in the system.

Exercise 21. In a certain system, a customer must first be served by server 1 and then by server 2. The service times at server i are exponential with rate μ_i, $i = 1, 2$. An arrival finding server 1 busy waits in line for that server. Upon completion of service at server 1, a customer either enters service with server 2 if that server is free or else remains with server 1 until server 2 is free. Customers depart the system after being served by server 2. Suppose that when you arrive there is one customer in the system and that customer is being served by server 1. What is the expected total time you spend in the system?

Exercise 22. Suppose in Exercise 21 you arrive to find two others in the system, one being served by server 1 and one by server 2. What is the expected time you spend in the system? Recall that if server 1 finishes before server 2, then server 1's customer will remain with him until server 2 becomes free.

Exercise 23. A flashlight needs two batteries to be operational. Consider such a flashlight along with a set of n functional batteries—battery 1, battery 2, ..., battery n. Initially, battery 1 and battery 2 are installed. Whenever a battery fails, it is immediately replaced by the lowest numbered functional battery that has not yet been put in use. Suppose that the lifetimes of

the different batteries are independent exponential random variables each having rate μ. At a random time, call it T, a battery will fail and our stockpile will be empty. At that moment exactly one of the batteries—which we call battery X—will not yet have failed.

(a) What is $P\{X = n\}$?

(b) What is $P\{X = 1\}$?

(c) What is $P\{X = i\}$?

(d) Find $E[T]$.

(e) What is the distribution of T?

Exercise 24. There are two servers available to process n jobs. Initially, each server begins work on a job. Whenever a server completes work on a job, that job leaves the system and the server begins processing a new job, provided there are still jobs waiting to be processed. Let T denote the time until all jobs have been processed. If the time that it takes server i to process a job is exponentially distributed with rate μ_i, $i = 1, 2$, find $E[T]$ and $\mathrm{Var}(T)$.

Exercise 25. Customers can be served by any of three servers, where the service times of server i are exponentially distributed with rate μ_i, $i = 1, 2, 3$. Whenever a server becomes free, the customer who has been waiting the longest begins service with that server.

(a) If you arrive to find all three servers busy and no one waiting, find the expected time until you depart the system.

(b) If you arrive to find all three servers busy and one person waiting, find the expected time until you depart the system.

Exercise 26. Each entering customer must be served first by server 1, then by server 2, and finally by server 3. The amount of time it takes to be served by server i is an exponential random variable with rate μ_i, $i = 1, 2, 3$. Suppose you enter the system when it contains a single customer who is being served by server 3.

(a) Find the probability that server 3 will still be busy when you move over to server 2.

(b) Find the probability that server 3 will still be busy when you move over to server 3.

(c) Find the expected amount of time that you spend in the system.

(d) Suppose that you enter the system when it contains a single customer who is being served by server 2. Find the expected amount of time that you spend in the system.

Exercise 34. Two individuals, A and B, both require kidney transplants. If she does not receive a new kidney, then A will die after an exponential time with rate μ_A, and B after an exponential time with rate μ_B. New kidneys arrive in accordance with a Poisson process having rate λ. It has been decided that the first kidney will go to A, or to B if B is alive and A is not at that time, and the next one to B if still living.

(a) What is the probability that A obtains a new kidney?

(b) What is the probability that B obtains a new kidney?

Exercise 35. Let $\{N(t) : t \geq 0\}$ be a right-continuous counting process with $N(0) = 0$, independent increments, and $N(s + t) - N(s) \sim \mathrm{Poisson}(\lambda t)$ for every $s, t \geq 0$. Show that its successive interarrival times are independent exponential random variables with rate λ.

Exercise 36. Let $S(t)$ denote the price of a security at time t. A popular model for the process $\{S(t), t \geq 0\}$ supposes that the price remains unchanged until a shock occurs, at which time the price is multiplied by a random factor. If we let $N(t)$ denote the number of shocks by time t, and let X_i denote the ith multiplicative factor, then this model supposes that

$$S(t) = S(0) \prod_{i=1}^{N(t)} X_i,$$

where $\prod_{i=1}^{N(t)} X_i$ is equal to 1 when $N(t) = 0$. Suppose that the X_i are independent exponential random variables with rate μ, that $\{N(t), t \geq 0\}$ is a Poisson process with rate λ, that $\{N(t), t \geq 0\}$ is independent of the X_i, and that $S(0) = s$.

(a) Find $E[S(t)]$.

(b) Find $E[S^2(t)]$.

Exercise 37. A machine works for an exponentially distributed time with rate μ and then fails. A repair crew checks the machine at times distributed according to a Poisson process with rate λ; if the machine is found to have failed then it is immediately replaced. Find the expected time between replacements of machines.

Exercise 38. Let $\{M_i(t), t \geq 0\}$, $i = 1, 2, 3$, be independent Poisson processes with respective rates λ_i, $i = 1, 2, 3$, and set

$$N_1(t) = M_1(t) + M_2(t), \qquad N_2(t) = M_2(t) + M_3(t).$$

The stochastic process $\{(N_1(t), N_2(t)), t \geq 0\}$ is called a bivariate Poisson process.

(a) Find $P\{N_1(t) = n, N_2(t) = m\}$.

(b) Find $\mathrm{Cov}(N_1(t), N_2(t))$.

Exercise 39. A certain scientific theory supposes that mistakes in cell division occur according to a Poisson process with rate 2.5 per year, and that an individual dies when 196 such mistakes have occurred. Assuming this theory, find

(a) the mean lifetime of an individual,

(b) the variance of the lifetime of an individual.

Also approximate

(c) the probability that an individual dies before age 67.2,

(d) the probability that an individual reaches age 90,

(e) the probability that an individual reaches age 100.

Exercise 40. Show that if $\{N_i(t), t \geq 0\}$ are independent Poisson processes with rate λ_i, $i = 1, 2$, then $\{N(t), t \geq 0\}$ is a Poisson process with rate $\lambda_1 + \lambda_2$, where

$$N(t) = N_1(t) + N_2(t).$$

Exercise 41. In Exercise 40, what is the probability that the first event of the combined process is from the N_1 process?

Exercise 42. Let $\{N(t), t \geq 0\}$ be a Poisson process with rate λ. Let S_n denote the time of the nth event. Find

(a) $E[S_4]$,

(b) $E[S_4 \mid N(1) = 2]$,

(c) $E[N(4) - N(2) \mid N(1) = 3]$.

Exercise 43. Customers arrive at a two-server service station according to a Poisson process with rate λ. Whenever a new customer arrives, any customer that is in the system immediately departs. A new arrival enters service first with server 1 and then with server 2. If the service times at the servers are independent exponentials with respective rates μ_1 and μ_2, what proportion of entering customers completes their service with server 2?

Exercise 44. Cars pass a certain street location according to a Poisson process with rate λ. A woman who wants to cross the street at that location waits until she can see that no cars will come by in the next T time units.

(a) Find the probability that her waiting time is 0.

(b) Find her expected waiting time.

Exercise 47. Consider a two-server parallel queueing system where customers arrive according to a Poisson process with rate λ, and where the service times are exponential with rate μ. Moreover, suppose that arrivals finding both servers busy immediately depart without receiving any service, whereas those finding at least one free server immediately enter service and then depart when their service is completed.

(a) If both servers are presently busy, find the expected time until the next customer enters the system.

(b) Starting empty, find the expected time until both servers are busy.

(c) Find the expected time between two successive lost customers.

Exercise 48. Consider an n-server parallel queueing system where customers arrive according to a Poisson process with rate λ, where the service times are exponential random variables with rate μ, and where any arrival finding all servers busy immediately departs without receiving any service. If an arrival finds all servers busy, find

(a) the expected number of busy servers found by the next arrival,

(b) the probability that the next arrival finds all servers free,

(c) the probability that the next arrival finds exactly i of the servers free.

Exercise 49. Events occur according to a Poisson process with rate λ. Each time an event occurs, we must decide whether or not to stop, with our objective being to stop at the last event to occur prior to some specified time T, where $T > 1/\lambda$. Consider the strategy that stops at the first event to occur after some fixed time s, $0 \leq s \leq T$.

(a) Using this strategy, what is the probability of winning?

(b) What value of s maximizes the probability of winning?

(c) Show that one's probability of winning when using the preceding strategy with the value of s specified in part (b) is $1/e$.

Exercise 50. The number of hours between successive train arrivals at the station is uniformly distributed on $(0, 1)$. Passengers arrive according to a Poisson process with rate 7 per hour.

Suppose a train has just left the station. Let X denote the number of people who get on the next train. Find

(a) $E[X]$,

(b) $\mathrm{Var}(X)$.

Exercise 51. If an individual has never had a previous automobile accident, then the probability he or she has an accident in the next h time units is $\beta h + o(h)$; on the other hand, if he or she has ever had a previous accident, then the probability is $\alpha h + o(h)$. Find the expected number of accidents an individual has by time t.

Exercise 52. Teams 1 and 2 are playing a match. The teams score points according to independent Poisson processes with respective rates λ_1 and λ_2. If the match ends when one of the teams has scored k more points than the other, find the probability that team 1 wins.

Exercise 53. The water level of a certain reservoir is depleted at a constant rate of 1000 units daily. The reservoir is refilled by randomly occurring rainfalls. Rainfalls occur according to a Poisson process with rate 0.2 per day. The amount of water added to the reservoir by a rainfall is 5000 units with probability 0.8 or 8000 units with probability 0.2. The present water level is just slightly below 5000 units.

(a) What is the probability the reservoir will be empty after five days?

(b) What is the probability the reservoir will be empty sometime within the next ten days?

Exercise 54. A viral linear DNA molecule of length 1 is often known to contain a certain marked position, with the exact location of this mark being unknown. One approach to locating the marked position is to cut the molecule by agents that break it at points chosen according to a Poisson process with rate λ. Letting m denote the location on the line of the marked position, if L_1 denotes the last Poisson event time before m and R_1 denotes the first Poisson event time after m, find

(a) $P\{L_1 = 0\}$,

(b) $P\{L_1 < x\}$, $0 < x < m$,

(c) $P\{R_1 = 1\}$,

(d) $P\{R_1 > x\}$, $m < x < 1$.

If the cutting procedure is utilized on n identical copies of the molecule, yielding the data L_i, R_i, $i = 1, \ldots, n$, then the marked position lies between L and R, where

$$L = \max_i L_i, \qquad R = \min_i R_i.$$

Find $E[R - L]$, and show that

$$E[R - L] \sim \frac{2}{n\lambda}.$$

Exercise 55. Consider a single server queueing system where customers arrive according to a Poisson process with rate λ, service times are exponential with rate μ, and customers are served in the order of their arrival. Suppose that a customer arrives and finds $n - 1$ others in the system. Let X denote the number in the system at the moment that customer departs. Find the probability mass function of X.

Exercise 56. An event independently occurs on each day with probability p. Let $N(n)$ denote the total number of events that occur on the first n days, and let T_r denote the day on which the rth event occurs.

(a) What is the distribution of $N(n)$?

(b) What is the distribution of T_1?

(c) What is the distribution of T_r?

(d) Given that $N(n) = r$, show that the set of r days on which events occurred has the same distribution as a random selection without replacement of r of the values $1, 2, \ldots, n$.

Exercise 57. Events occur according to a Poisson process with rate $\lambda = 2$ per hour.

(a) What is the probability that no event occurs between 8 P.M. and 9 P.M.?

(b) Starting at noon, what is the expected time at which the fourth event occurs?

(c) What is the probability that two or more events occur between 6 P.M. and 8 P.M.?

Exercise 58. Consider the coupon collecting problem where there are m distinct types of coupons, and each new coupon collected is type j with probability p_j, where

$$\sum_{j=1}^{m} p_j = 1.$$

Suppose you stop collecting when you have a complete set of at least one of each type. Show that

$$P\{i \text{ is the last type collected}\} = E\left[\prod_{j \neq i}\left(1 - U^{p_j/p_i}\right)\right],$$

where U is a uniform random variable on $(0, 1)$.

Exercise 59. There are two types of claims that are made to an insurance company. Let $N_i(t)$ denote the number of type i claims made by time t, and suppose that $\{N_1(t), t \geq 0\}$ and $\{N_2(t), t \geq 0\}$ are independent Poisson processes with rates $\lambda_1 = 10$ and $\lambda_2 = 1$. The amounts of successive type 1 claims are independent exponential random variables with mean \$1000, whereas the amounts from type 2 claims are independent exponential random variables with mean \$5000. A claim for \$4000 has just been received; what is the probability it is a type 1 claim?

Exercise 60. Customers arrive at a bank at a Poisson rate λ. Suppose two customers arrived during the first hour. What is the probability that

(a) both arrived during the first 20 minutes?

(b) at least one arrived during the first 20 minutes?

Exercise 61. A system has a random number of flaws that we will suppose is Poisson distributed with mean c. Each of these flaws will, independently, cause the system to fail at a random time having distribution G. When a system failure occurs, suppose that the flaw causing the failure is immediately located and fixed.

(a) What is the distribution of the number of failures by time t?

(b) What is the distribution of the number of flaws that remain in the system at time t?

(c) Are the random variables in parts (a) and (b) dependent or independent?

Exercise 62. Suppose that the number of typographical errors in a new text is Poisson distributed with mean λ. Two proofreaders independently read the text. Suppose that each error

is independently found by proofreader i with probability p_i, $i = 1, 2$. Let X_1 denote the number of errors that are found by proofreader 1 but not by proofreader 2. Let X_2 denote the number of errors that are found by proofreader 2 but not by proofreader 1. Let X_3 denote the number of errors that are found by both proofreaders. Finally, let X_4 denote the number of errors found by neither proofreader.

(a) Describe the joint probability distribution of X_1, X_2, X_3, X_4.

(b) Show that
$$\frac{E[X_1]}{E[X_3]} = \frac{1-p_2}{p_2} \qquad \text{and} \qquad \frac{E[X_2]}{E[X_3]} = \frac{1-p_1}{p_1}.$$

(c) Suppose now that λ, p_1, and p_2 are all unknown. By using X_i as an estimator of $E[X_i]$, $i = 1, 2, 3$, present estimators of p_1, p_2, and λ.

(d) Give an estimator of X_4, the number of errors not found by either proofreader.

Exercise 63. Consider an infinite server queueing system in which customers arrive in accordance with a Poisson process with rate λ, and where the service distribution is exponential with rate μ. Let $X(t)$ denote the number of customers in the system at time t. Find

(a) $E[X(t+s) \mid X(s) = n]$,

(b) $\mathrm{Var}[X(t+s) \mid X(s) = n]$.

Also, consider an infinite server queueing system in which customers arrive according to a Poisson process with rate λ, and where the service times are all exponential random variables with rate μ. If there is currently a single customer in the system, find the probability that the system becomes empty when that customer departs.

Exercise 64. Suppose that people arrive at a bus stop in accordance with a Poisson process with rate λ. The bus departs at time t. Let X denote the total amount of waiting time of all those who get on the bus at time t. We want to determine $\mathrm{Var}(X)$. Let $N(t)$ denote the number of arrivals by time t.

(a) What is $E[X \mid N(t)]$?

(b) Argue that
$$\mathrm{Var}[X \mid N(t)] = \frac{N(t)t^2}{12}.$$

(c) What is $\mathrm{Var}(X)$?

Exercise 65. An average of 500 people pass the California bar exam each year. A California lawyer practices law, on average, for 30 years. Assuming these numbers remain steady, how many lawyers would you expect California to have in 2050?

Exercise 66. Policyholders of a certain insurance company have accidents at times distributed according to a Poisson process with rate λ. The amount of time from when the accident occurs until a claim is made has distribution G.

(a) Find the probability there are exactly n incurred but as yet unreported claims at time t.

(b) Suppose that each claim amount has distribution F, and that the claim amount is independent of the time that it takes to report the claim. Find the expected value of the sum of all incurred but as yet unreported claims at time t.